# Corner payments and a dimension bound of 57 for finite-volume hyperbolic Coxeter polytopes

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**Abstract**

We prove a dimension bound of57 for finite-sided, finite-volume real hyperbolic Coxeter polytopes. The argument assigns nonnegative weights to corners of two-dimensional faces. Triangles and quadrilaterals require enough total weight, while the finite and affine Coxeter diagrams at a vertex limit how much weight can be spent there. A global incidence inequality makes these requirements incompatible in dimension 58. The finite-vertex upper bound is exactly 377, the curvature coefficient; ideal vertices retain surplus at least 36, and the two global extrema supply a further correction 4205. Exact finite certificates support the local rule and the vertex estimates. We describe their algorithms and the distinction between geometric coverage and arithmetic verification. The treatment includes nonsimple ideal vertices, whose links are products of simplices. A separate appendix proves the higher-dimensional exclusion, including the unbounded range, without assuming that a facet is Coxeter.

## 1 Statement and proof architecture

**Theorem 1.** *Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most 57.*

No arithmeticity, compactness, simple-cusp or facet-count hypothesis is imposed. The theorem belongs to the combinatorial approach to dimension bounds developed by Vinberg, Prokhorov and Khovanskii: small faces force local contributions that a high-dimensional vertex cannot accommodate [1, 3]. The largest presently known dimension of a finite-volume example is 21; Borcherds constructs an arithmetic reflection group in that dimension [2]. Thus the present bound still leaves a substantial existence gap.

This is a computer-assisted argument. The exact endpoint is `fresh_d13_context_dimension57_20261018`; its recursive source manifest identifies the immutable proof atlas.

The main argument excludes dimension58. Appendices D and E treat every higher dimension: first by a finite set of payment families, and then by an all-rank narrow-facet argument. We do not infer this tail from the single excluded dimension, or pass to a facet whose angles have not been proved Coxeter. The marked local rules and their exact arithmetic certificates are internal parts of the argument, not additional published assumptions.

For a two-face$F$ with$s(F)$ sides, write

$$\mathcal{D}_5(P) = \sum_F (5 - s(F)).$$

A simultaneous nonnegative corner rule paying at least 2 on each triangle and 1 on each quadrilateral gives an upper bound for$\mathcal{D}_5$. The final rule has finite-vertex charge at most 377. At an ideal vertex

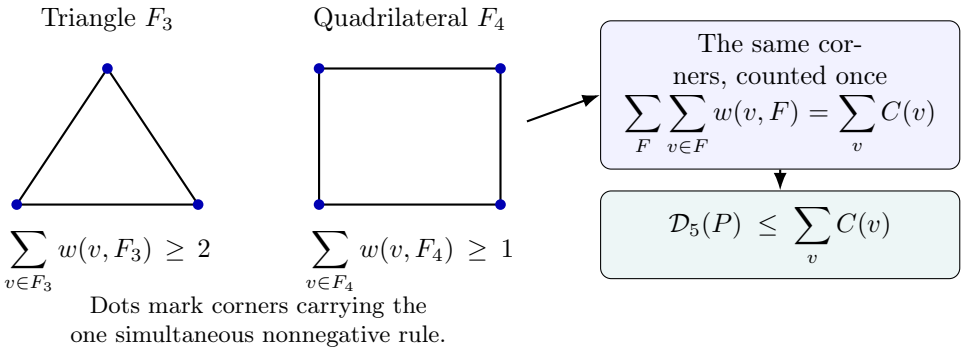



Figure 1: Corner accounting in Lemma 2. The polygons are schematic, not metric Coxeter realizations. No equal individual weights are implied. Nonnegativity handles faces with at least five sides; the incidence argument supplies the separate lower bound.

its actual charge$C_{\text{act}}$ satisfies$\kappa - C_{\text{act}} \geq 36$, where$\kappa$ is the link shape curvature. The global lower bound is

$$\mathcal{D}_5 \geq 377v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

The contradiction is therefore

$$0 \geq 4205 + 36v_i > 0.$$

The remainder proves the local and arithmetic interfaces that make this subtraction valid.

### 1.1 The common counting argument

An angle in this paper is a corner $(v, F)$ of an actual two-face. A payment is a nonnegative weight $w(v, F)$. Write $C(v) = \sum_{F\ni v} w(v, F)$ for its total charge at a vertex. The elementary local-to-global step is worth stating independently of the particular rule and of the certificate files.

**Lemma 2** (Payment comparison)**.** *Suppose every triangular face receives total payment at least 2 and every quadrilateral receives at least 1. Then*

$$\mathcal{D}_5(P) \leq \sum_v C(v).$$

*More generally, suppose an incidence argument gives*

$$\mathcal{D}_5(P) \geq Bv_f + \sum_{v \text{ ideal}} \kappa(v) + \Delta, \qquad \Delta > 0,$$

*while a simultaneous payment rule has $C(v) \leq B$ at every finite vertex and $C(v) \leq \kappa(v) - \varepsilon$ at every ideal vertex, where $\varepsilon \geq 0$. These hypotheses are inconsistent.*

*Proof.* A face with $s$ sides needs payment at least $5 - s$. This is2 for a triangle and 1 for a quadrilateral. For $s \geq 5$ it follows from nonnegativity. Sum over faces and then over their corners. Combining the resulting upper bound with the proposed lower bound gives $0 \geq \Delta + \varepsilon v_i > 0$. □

The geometry enters in choosing a rule that works at every actual face. The computation enters after that reduction, when the possible vertex diagrams and their contextual costs have been specified. The same corner may not be spent twice merely because two local arguments could have

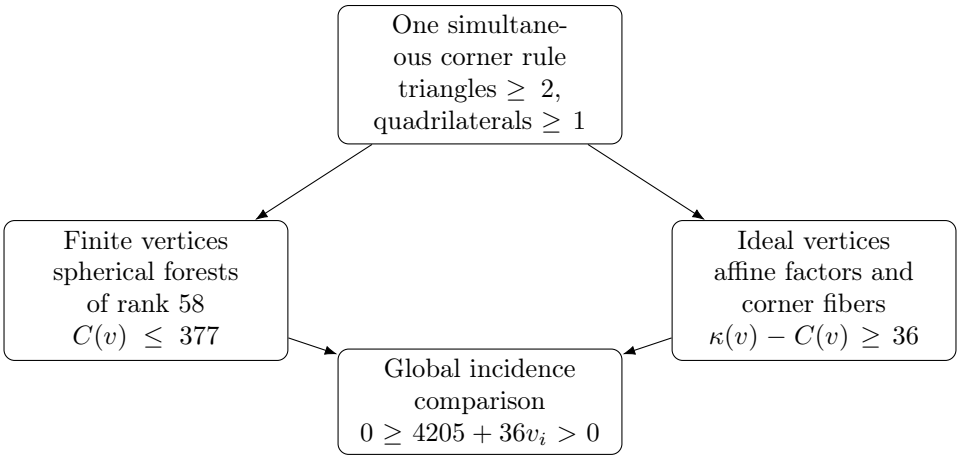



Figure 2: The common geometric rule is established before the finite and ideal arithmetic separate. The strict correction4205 comes from the two extrema of a generic linear function on the polytope.

used it: all priorities and compensations must belong to one simultaneous rule. Appendix B gives that rule explicitly, as a function of a marked whole vertex diagram. For the preliminary finite count we use the same tables with the stated $B_{13}$ endpoint restored and without the final three rank-15-guarded changes. We call this preliminary function the preceding rule below.

The proof first separates into finite and ideal vertices. At a finite vertex we maximize the rule over spherical forests of total rank 58. At an ideal vertex we bound each affine factor, multiply by its actual corner-fiber multiplicity, and compare with the curvature of the product link. These branches reunite in the payment comparison. The final changes to the rule use whole components: a visible path inside a larger component does not carry the same contextual bonus. Figure 2 records this division.

## 2 Actual corners, whole components and the previous finite count

We use outward norm-two roots, with diagonal Gram entries2 and nonpositive off-diagonal entries. At an actual two-face in dimension$n$, the supporting white roots form a positive-definite rank$n-2$ diagram. White roots together with one black side give a strictly positive-definite edge diagram; adjacent corners are positive semidefinite. A finite corner has an elliptic forest of rank$n$.

These assertions refer to actual faces of the original chamber. We record two elementary feasibility arguments that will also control additional supporting roots. Let $p$ be a strict interior timelike point. A positive-definite principal Gram matrix $B$ has nonpositive off-diagonal entries and an entrywise nonnegative inverse. If $b_i = -(p, e_i) > 0$, then

$$q = p + \sum_i (B^{-1}b)_i e_i$$

is orthogonal to the selected roots and has $q^2 = p^2 - b^T B^{-1} b < 0$. It is future pointing and strict against every other chamber wall. Thus a positive principal set has an actual finite common point, rather than merely a formal Gram model.

An irreducible affine principal set has a positive kernel vector $(u_i)$. Its root combination $\nu = \sum_i u_i e_i$ is null and nonzero, since $(p, \nu) < 0$. It is feasible against every chamber wall. Its projective point is an ideal vertex: an ideal-sphere point is extreme in the closed Klein ball and hence in

the chamber. The affine diagram is a whole factor of that cusp, because every proper principal subdiagram of an irreducible affine diagram is positive definite. This is also the containment input of [7, Section 1].

**Lemma 3** (Opposite sides and minimal cores)**.** *For either opposite pair of sides of an actual quadrilateral, the supporting whites and that pair form an indefinite principal set. An inclusion-minimal indefinite subset contains both marked sides. Its order is at most ten; if all proper principal subsets are elliptic, its order is at most five.*

*Proof.* A positive set would have an actual common finite point by the projection argument. A positive-semidefinite singular set would have a common ideal point by its affine kernel. Opposite closed sides of a nondegenerate quadrilateral are disjoint, so neither is possible. Deleting either marked side leaves a subset of an actual finite edge diagram. Hence both marks lie in every minimal indefinite subset. The published Lannér/quasi-Lannér order bounds now apply; the classification through order ten is [4]. □

Two such opposite cores cannot be disjoint and orthogonal, since that would give two negative directions. Their overlap or a joining path supplies an adjacent-side route. The local theorem must then restore all the other supporting whites; an argument on the displayed cores alone would not justify a whole-component payment.

A whole component is a connected component of the full actual corner diagram, not merely of a displayed core. Presence conditions in a weight refer only to such whole components. This distinction is essential when additional supporting whites are restored.

**Lemma 4** (Two ways to seal a component)**.** *Let $G$ be the Gram matrix of an actual indefinite core.*

1. *A Gram-kernel vector $z$ of this core gives the actual zero root relation $\sum_i z_i e_i = 0$. Every further wall therefore satisfies $\sum_i z_i(e, e_i) = 0$.*
2. *Suppose $G$ is nonsingular, an added forest contacts it only at a position $p$, and $(G^{-1})_{pp} = r > 1$. If every connected component of the added forest contacts $p$, its total rank is at most $1/(r-1)$.*

*Proof.* The span of an indefinite core contains a timelike vector. The root combination associated with a Gram-kernel vector is null and orthogonal to that timelike vector. The perpendicular space is positive definite, so the combination is zero. This differs from the nonzero feasible null vector of an affine block.

For the second statement let $B$ be the positive Gram matrix of all added supporting whites, including indirect tails, and let $u$ be their contact column at $p$. Schur elimination and the one-negative- direction condition give $B - ruu^T \succeq 0$, whence $u^T B^{-1} u \leq 1/r$.

For a connected square-two finite Coxeter Gram of rank $k$, every inverse diagonal is at least $k/(k+1)$. This follows by induction: delete the chosen node, let the remaining component ranks be $k_j$, and use a contact of magnitude at least one into each component. Inverse positivity gives Schur cost at least $\sum_j k_j/(k_j+1) \geq (k-1)/k$. The resulting inverse diagonal is at least $1/(2-(k-1)/k) = k/(k+1)$; the rank-one base is $1/2$. For an added forest of total rank $M$ contacted in each component, the same bound gives $u^T B^{-1} u \geq M/(M+1)$. Thus $M/(M+1) \leq 1/r$, proving $M \leq 1/(r-1)$. □

The second argument takes all roots of the growing forest at once. A bound on its immediate neighbors would leave indirect extensions uncontrolled. For a radical seal, nonpositive contact signs and the displayed zero relation restrict every further root; the local short-path alternatives then

force either a persistent payment or orthogonality. An indirect extension has a first contacting root, so this is the step that turns a core component into a whole one.

At an ideal vertex with affine factor ranks$r_i$, the link is $\prod_i \Delta^{r_i}$, with$\sum r_i = n-1$. A marked pair in factor$i$ occurs in$\prod_{j\neq i}(r_j+1)$ actual corner fibers. Hence an affine factor envelope$(r+1)c(r)$ gives

$$C_{\mathrm{act}} \le C_{\mathrm{up}} = V\sum_i c(r_i), \qquad V=\prod_i (r_i+1). \tag{1}$$

The factor-fiber theorem includes nonsimple ideal vertices. Whole affine factors used locally are established by the positive kernel and actual corner semidefiniteness, before applying this multiplicity. Indeed an edge of the product link varies in just one simplex factor. A fixed pair of its facet labels determines an edge in that factor and an arbitrary vertex in every other factor. These are precisely the $V/(r_i+1)$ fibers. The marked pair restores the varying affine factor; all other factors appear with one node deleted. Summing an unweighted copy of each affine table would miss this multiplicity.

## 2.1 The finite forest algorithm

For the finite calculation multiply charges by64. Its alphabet is

$$\{A_r : 1\le r\le 58\}\cup\{B_r : 2\le r\le 58\}\cup\{D_r : 4\le r\le 58\}\cup\{E_6,E_7,E_8,F_4,H_3,H_4,I_2^*\}.$$

The rank-two symbol $I_2^*$ represents all remaining $I_2(m)$, $m\ge 5$, in the cost calculation. The local all-label theorem gives their common relevant charge and interactions; this notation is not a finite cutoff on $m$ or an identification of their metrics. An admissible formal forest has multiplicities $m_T\ge 0$ with $\sum_T \operatorname{rank}(T)m_T = 58$. Actual finite vertices are among these forests. The calculation needs no converse realization theorem.

The ordinary cost consists of a component table $b(T)$ and a symmetric interaction table $j(T,U)$; context bonuses are separate. In particular $j(T,T)$ concerns two distinct copies of $T$. The cost is

$$\begin{aligned} C_{64}(m) = &\sum_T b(T)m_T + \sum_T j(T,T)\binom{m_T}{2} \\ &+ \sum_{T<U} j(T,U)m_Tm_U + \Psi(m). \end{aligned} \tag{2}$$

The tables and the context function belong to the simultaneous local rule. Their identification with actual corner charges is a geometric obligation, separate from maximizing this expression. Appendix A supplies the actual-face coverage and triangle arguments. Appendix C specifies the complete context function and the finite refinement used in this calculation.

Fix all context-sensitive counts before the residual calculation. After this split the residual alphabet $\mathcal{U}$ has positive integer ranks $r(T)$ and additive integer weights $w(T)$, with no remaining pair interaction. For a residual rank $R\le 58$ define

$$V(0)=0, \qquad V(s)=\max_{\substack{T\in\mathcal{U},r(T)\le s\\ s-r(T)\text{ reachable}}}\{V(s-r(T))+w(T)\} \quad (1\le s\le R). \tag{3}$$

An empty maximum is an unreachable state, not an ordinary integer. Evaluate in increasing rank, recording a last item for each finite maximum.

**Proposition 5** (Residual certificate)**.** *The algorithm* (3) *terminates and computes the exact maximum over residual multisets. A saved table certifies the same claim if it has base value zero, dominates every allowed last-item transition, and has a tight last-item witness at every positive rank carrying a finite stored value. At positive ranks, a state is unreachable exactly when it has no reachable last-item predecessor; rank zero is the reachable base.*

*Proof.* There are at most 59 ranks and finitely many transitions at each rank. Every predecessor has smaller rank. By induction, removing one component from any nonempty multiset bounds its weight by one of the candidates in (3). Conversely, append the chosen component to an attaining predecessor multiset. This proves both inequalities and reachability. The same induction applies to a saved table using all its transition inequalities and tight witnesses at every stored finite value. Positive item ranks make each tight predecessor smaller, so strong rank induction rules out spurious finite entries. Following witnesses reaches zero after at most 58 steps. Checking only an attaining path would not bound omitted multisets. □

In the actual context split, fixed representative counts have $\sum_{r=9}^{17} rc_r \le 58$, with a separate $A_{12}/D_{12}$ split, fixed $A_{17}$, and $0 \le k_5 \le \lfloor (58 - \sum rc_r)/5 \rfloor$. The residual ranks are $1, 2, 3, 4, 6, 7, 8$ and 17 through 58. Expand every class that can exceed 377 into all actual-type fibers. Within each fiber fix the complete $A_1, A_2$ counts. For the only remaining presence guard, split into no $A_3$ or reserve one $A_3$ and allow further copies without repeating its $D_{14}$ bonus. The two exceptional increasing families are bounded separately, rather than omitted from the context cover. All count ranges are bounded by rank 58. These facts prove termination of the full context algorithm; the geometric domination theorem establishes that its cover includes every actual forest.

**Proposition 6** (Computed finite interface)**.** *The preceding simultaneous rule in ambient 58 pays every triangle at least2 and every quadrilateral at least 1. Every actual rank 58 finite forest has charge at most 377 except precisely the following eight formal forests: one$B_{13}$ or$D_{13}$ together with any multiset of three $A_{15}$ or$D_{15}$ components. Each of those eight has charge378. All ideal weights admit the envelope of Section 5.*

*Certificate and completeness argument.* The frozen coupled count starts from the complete 3298 fixed-vector/rank 5 cover, retaining fixed$A_{17}$ separately from residual$D_{17}$ and separating the$A/B_{12}$ and$D_{12}$ interaction types. Simultaneous actual base and cross-cost domination covers exceptional finite types and the arbitrary-label rank-two family. Its two potentially increasing context families are bounded separately:759 rows for the$A_1/D_5$ upgrade, and 539 rows for the$A_2/A_4$ guarded family, with all 14 costly rows refined through 77 actual assignments. Both bounds are strictly below377.

The complementary complete 39 costly classes have748 fixed actual-type fibers. All actual$A_1, A_2$ multiplicities are fixed explicitly. The remaining$A_3$ presence guard is split into no$A_3$ and one reserved$A_3$; the latter may contain more$A_3$ without multiplying the presence bonus. All other context guards are fixed, and residual pair interactions vanish. The1549 fresh residual tables certify every allowed last-item inequality, unreachable rank and attaining predecessor. Induction on an item's removal therefore proves their upper bounds for arbitrary residual multisets. Reconstructing every winning forest verifies its full contextual cost. All38 other classes are at most 377; the residual-zero rank 13-plus-three15 class gives exactly the eight listed forests. Their charge is$78 + 3(100)$.

The complete source, saved tables and separate replay are in `fresh_d17_a16_coupled_ambient58_frontier_20261018`. This proof interface is not inferred from sampled maximizing forests. The written universal local atlas is also required; the Bellman certificate alone does not prove that its costs correspond to valid corner payments. □

A separate frozen$B_{13}$ endpoint theorem removes pair$(0, 12)$, with no new coupon. Its complete selected domain has no affected image, but that absence alone is insufficient: two older affine-$B_{13}$ singleton routes really consumed the endpoint unit. Their whole cores instead give distinct already-unit$A_1/D_{12}$ short-leaf contributions. The other endpoint-like pair$(1, 12)$ needed by the current$D_{12}$ theorem is retained. Thus the four$B_{13}$ forests in Proposition 6 drop to 377.

# 3 The dimension-specific final local rule

**Theorem 7** (Three-rank 15 transfer)**.** *In ambient dimension 58, make these changes to the simultaneous rule:*

1. *In a whole finite$D_{13}$, give pairs$(0, 11), (0, 12)$ weight$1/2$ instead of1 only if its actual corner has three distinct whole unmarked $A_{15}$ or$D_{15}$ components.*
2. *Add$1/2$ to whole$D_{28}$ pair$(0, 12)$ only if its corner has two such whole unmarked rank 15 components.*
3. *Add$1/2$ between an endpoint of whole$A_{12}$ and the long endpoint of whole$D_{16}$ under the same two-component presence guard.*

*All ideal weights and other finite weights remain. Every triangle still pays at least 2 and every quadrilateral at least 1.*

Witnesses are not counted with multiplicity. The negative guard uses $13+3\cdot15 = 58$ ranks, so the corner has exactly those four components. The two positive guards similarly force exactly$D_{28}+2\cdot15$ or $A_{12} + D_{16} + 2 \cdot 15$. The theorem is not asserted in larger ambient dimensions.

## 3.1 A general inverse bound for attached forests

**Lemma 8.** *For a connected positive-definite Coxeter Gram matrix$B$ of rank$k$, with diagonal2 and every nonzero native magnitude at least 1, $(B^{-1})_{ii} \geq k/(k + 1)$ for every$i$. If a finite forest of total rank$M$ has a nonnegative contact vector$u$ with at least one unit contact in every component, then$u^t B^{-1}u \geq M/(M + 1)$.*

*Proof.* Inverse matrices of these positive-definite symmetric$Z$-matrices are entrywise nonnegative. Delete the chosen root; let the remaining connected components have ranks$k_j$. Every component has a contact with the root. By induction and inverse positivity its contribution to the root's Schur projection is at least$k_j/(k_j + 1)$. Since

$$\sum_j \frac{k_j}{k_j + 1} \geq \frac{\sum_j k_j}{1 + \sum_j k_j} = \frac{k - 1}{k},$$

Schur complementation gives $(B^{-1})_{ii} \geq (2 - (k - 1)/k)^{-1} = k/(k + 1)$. The rank-one base is$1/2$. Larger native magnitudes or multiple contacts only increase the lower bound. Apply it in every contacted component of a forest and use the same sum inequality to prove the second assertion. □

## 3.2 Complete selected cases and arbitrary additional whites

The fresh complete target domain contains801 patterns,862$D_{13}$ maps, 33 affected contexts and 76 affected pair maps. The physical cores have norm-two roots; full Lorentz, white and all four

edge signatures and actual corner classifications are reconstructed exactly. The native-domain completeness is inherited from the universal reduction, rather than claimed from these records alone.

There are five proper completions. Their missing fork leaf is genuinely new: old corner roots outside the displayed$A_{12}$ are orthogonal to it. Its forced unit contact has inverse projection square6,12,30 or102, exceeding2. Additional same-end contacts cannot lower the bound. Mixed-end contacts give an independently paid short path. Thus these completions are impossible.

Four radical cores, numbered1985,2378,2575,2581, seal the entire displayed core. Indeed the whole selected$D_{13}$ forbids contacts on its nodes; every remaining contact position has a nonzero radical coefficient. A new white must pair to zero with the radical. Nonempty contacts therefore meet both signs and give a retained adjacent-black shortcut of length at most8. Otherwise all contacts vanish and the unmarked$A_4$ is whole. It cannot coexist with the three rank 15 witnesses.

Twelve contexts similarly have a small displayed block with no permitted new contact of projection at most 2. It is whole: any indirect extension would have a first new root contacting it. The rank-capacity guard again excludes it.

In the final twelve contexts the old small factor$K$ has rank$s = 1, 2, 3$. Its unique permitted contact position$p$ has core-inverse square $r = 4/3, 3/2, 2$, respectively. The native gap excludes magnitudes greater than1; positive-end inverse bounds exclude two same-end contacts. Four cases also permit an opposite contact position$q$, with $J_{pp} = J_{qq} = 4/3$ and$J_{pq} = -4/3$. Two new roots using$p, q$ have residual squares$2/3$, so their full product is at most$-4/3 + 2/3 = -2/3$. Their forced native bond supplies the retained short path. Thus, without that escape, all new whites in a growing$K$ component contact the old core only at$p$; some may be orthogonal to it.

Let$Y$ contain all new whites in this whole component, including every indirect tail, and let$M$ be its rank. Every component of$Y$ contacts$K$. With contact vector$u$, the full Lorentz condition gives the positive semidefinite residual Schur matrix

$$G_Y - ruu^t \succeq 0, \qquad u^t G_Y^{-1} u \le 1/r.$$

Lemma 8 implies$rM/(M+1) \le 1$, hence $M \le 3, 2, 1$ respectively. In all cases the whole$K$ component has rank at most 4, contradicting the guard. This controls arbitrary supporting whites, not just immediate neighboring roots.

### 3.3 The universal affine-D12 consumer

The old opposite-fork-leaf affine-$D_{12}$ theorem genuinely used the newly lowered whole finite$D_{13}$ unit. It must be repaired separately. Number its affine factor$F$ of order13 with spine0 through 8, short leaves 9,10 at0 and 11,12 at8. Choose ideal marks$a = 9, b = 11$ and opposite black roots$c, d$. Let$Z$ be the elliptic spectator forest, orthogonal to$F$, and$x, y$ the nonnegative contacts of$c, d$ with$Z$.

An adjacent-black bond or a shared spectator neighbor is an independently paid short path. Otherwise$x, y$ have disjoint supports. The complete all-label end-profile reduction gives each end cost at most 8; a nonextreme end has cost at most 4. The exact Schur identity is

$$\frac{\det S}{h^2 k^2} = 1 - \frac{g_c + g_d}{12} + \frac{1}{12}(x/h - y/k)^t G_Z^{-1}(x/h - y/k).$$

Its determinant vanishes at the actual four-black star.

If an end is nonextreme, equality forces costs8 and 4 and$x = y = 0$. A quiet root attached to the selected old leaf gives the retained whole $A_1/D_{12}$ short-leaf unit. If it attaches to the mate, it gives whole $D_{13}$. Without the new guard its unit is unchanged. With the guard, $Z$ consists of exactly three orthogonal whole rank 15 factors. The three nonextreme equality profiles give, at a distinct

corner, respectively a retained$A_1/D_{12}$ unit, an independent whole affine $B_{13}$ priority, or a second guarded$D_{13}$ half. These exhaust24 physical cases. The current$B_{13}$ endpoint repair uses its$D_{12}$ cross unit, not a newly financed$D_{13}$ route; there is no circularity.

When both ends are extreme,$h = k = 1/2$, and the determinant becomes

$$(x - y)^t G_Z^{-1}(x - y) = 1. \tag{4}$$

The old two-nonempty-support theorem gives retained short paths or small sealed alternatives. The changed$D_{13}$ output belongs to the arbitrary one-sided branch, say$y = 0$. The quiet corner excludes$c$ and retains$Z$ orthogonally. If its$D_{13}$ guard holds,$Z$ is exactly three whole $A_{15}/D_{15}$ factors.

**Lemma 9.** *Under the strict edge conditions and disjoint-support hypothesis, energy1 on three whole$A_{15}/D_{15}$ factors forces a single unit contact at the long endpoint of one$D_{15}$, by one black root. The other two factors have zero contact with both black roots.*

*Proof.* For $A_{15}$ the inverse diagonal entries below2 occur only at nodes 0,1,13,14. They are 15/16, 7/4, 7/4, and 15/16, respectively. Two native contacts give energy at least 2 and violate strict edge positivity. The complete profiles are simple contacts at these four nodes or doubled contacts at0,14. The next native squared magnitude$(3 + \sqrt{5})/2$ is already excluded. For$D_{15}$ only its long endpoint has inverse diagonal below2, equal to 1, and only its simple contact fits.

Every allowed nonzero component energy is at least15/16; no$A_{15}$ profile has energy1. The finite exact inverse table also checks every disjoint two-sided profile difference: all are at least15/16 and none is1. Equal-node contacts are forbidden by disjoint support. Two active components would have total at least$15/8 > 1$. Equation (4) therefore forces precisely the stated$D_{15}$ contact. □

The two inactive components are orthogonal to the entire star and the active component. They remain whole, without indirect bridges. If$c$ attaches to selected$a$, the opposite corner is whole$A_{12}$ at an endpoint and whole$D_{16}$ at its long endpoint. If it attaches to$a$'s mate, that corner is whole$D_{28}$ at$(0, 12)$. Both physical30-root models have signature$(28, 1, 1)$, positive white rank 26, positive edge rank 27 and semidefinite corners. In either case the new half, guarded by the two inactive rank 15 factors, supplements the quiet$D_{13}$ half at a different actual corner. This proves the required total1.

### 3.4 Chosen-output and triangle retention

The previous selected and nonselected routes must retain their chosen positive contributions, including every later replacement route. The low66 contexts have order bound at most 12. The74 protected and 9 adaptive paths have length9, whereas the changed$D_{13}$ role has distance11. The current finite6/7/8 and cycle/tree routes terminate at short paths, protected leaves, small whole factors or unchanged ideal contributions.

The older affine-$D_{11}$ repairs use$D_{13}(1, 11)/(1, 12)$ units, retained here. The current center/leaf theorem uses$D_{13}(0, 10)$, at its fork center. The coupled order11 unmixed boundary gives$D_{11+r}$ marks $(r, n - 2)/(r, n - 1)$ of distance9; for$r = 2$ these are disjoint$D_{13}$ roles. The other current finite and contextual outputs are similarly retained. The new producer reconstructs all 216 literal configurations,564 later corners,24$D_{14}$ variants,332 positive$D_{12}$ outputs and the newer sealed families. These finite checks support the named interfaces; universal completeness remains the written routing theorem. The genuine unstored affine-$D_{12}$ exception was repaired above.

For triangles use the complete actual-face theorem, not three pairs in one diagram. Lemma 17 first excludes the three reduced small affine patterns. Low cores give two distinct corner paths of length at most 5. The1025 high-core triples give paths of length at most 7 or one of four protected$E_8$ alternatives, with finite marked length6 or7 or whole exceptional affine order at most 9. Current

finite6 deletions retain all three exceptional length6 alternatives through their nearby whole leaf; the length7 alternative and other persistent-feature outputs survive. The present distance11 reduction changes none of these chosen payments. Ideal contributions are unchanged. This completes Theorem 7.

## 4 The new complete finite maximum

**Proposition 10.** *Every actual finite rank 58 forest under the final simultaneous rule has charge at most 377. The exact formal maximum is377.*

*Proof.* The$B_{13}$ theorem lowers four forests in Proposition 6 from 378 to 377. The new$D_{13}$ guard lowers the other four by1. Every other forest was already at most 377, except that the two new positive coupon families must be checked.

Rank capacity makes those families exact. They have respectively

$$170 + 2(100) + \tfrac{1}{2} = 370.5, \qquad 66 + 108.5 + 2(100) + 1 = 375.5.$$

The second total counts both$A_{12}$ endpoint coupons. The previous $A_{12}/D_{16}$ cross cost is zero. Each family has three possible multisets of two$A_{15}/D_{15}$ factors, so all six actual forests are covered. No additional context witness can fit. Outside them the change is pointwise nonincreasing. Thus the bound377 holds everywhere, and the eight changed forests attain it formally. A geometric realization of an attainer is neither required nor asserted. □

## 5 Positive ideal surplus

For affine ranks$r_i$ summing to 57 set

$$H(t) = \prod_i (1 + t + \cdots + t^{r_i}) = \sum_{j=0}^{57} h_j t^j, \qquad V = \prod_i (r_i + 1),$$
$$G = \sum_j h_j \min(j, 57 - j), \qquad \kappa = \frac{5G - 57V}{2}.$$

The geometric all-family envelope is unchanged by every final finite transfer. Its complete numerical table is

$$c(r) = r/2 \quad (1 \le r \le 8), \qquad c(r) = \frac{5r + 56}{r + 1} \quad (24 \le r \le 57),$$

together with the following entries:

| $r$ | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|
| $c(r)$ | 71/16 | 53/11 | 31/6 | 11/2 | 40/7 | 367/60 | 411/64 | 227/34 |
| $r$ | 17 | 18 | 19 | 20 | 21 | 22 | 23 | |
| $c(r)$ | 119/18 | 128/19 | 137/20 | 146/21 | 155/22 | 163/23 | 85/12 | |

The finite arithmetic below assumes the proved factor-charge envelope (1); polynomial evaluation does not establish its geometric meaning.

### 5.1 The terminal-domain algorithm

Let $\mathcal{P}(n,\ell)$ be the list of nondecreasing partitions of $n$ with parts at least $\ell$. Generate it by

$$\mathcal{P}(0,\ell)=\{()\},\qquad \mathcal{P}(n,\ell)=\bigcup_{q=\ell}^{n}\{(q,p):p\in\mathcal{P}(n-q,q)\}\quad (n>0).$$

The union is empty if $\ell>n$. The terminal cusp domain is

$$\mathcal{T}_{57}=\bigcup_{\ell=0}^{7}\{(\ell,p):p\in\mathcal{P}(57-\ell,8)\},\tag{5}$$

where a zero part is omitted. For each profile compute its polynomial by repeated convolution: start with $h_0^{(0)}=1$, and when adjoining rank $r$ set

$$h_j^{(k+1)}=\sum_{a=0}^{r}h_{j-a}^{(k)},$$

with out-of-range entries zero. Then compute $V,G,\kappa$ from their displayed definitions and set $m=\kappa/V-\sum_i c(r_i)$.

**Proposition 11** (Complete terminal calculation)**.** *The partition and convolution algorithms terminate, list every profile in* (5) *exactly once, and compute its margin by exact rational arithmetic. Their recorded complete output is2552 profiles, with*

$$\min_{\mathcal{T}_{57}} m=\frac{87}{10240}>0,\qquad \min_{\mathcal{T}_{57}} Vm=36.$$

*Algorithm correctness and computed output.* Every recursive call lowers $n$ by a positive integer, and every loop is finite. The unique first part of a sorted partition and its unique tail prove completeness and absence of duplicates by induction on $n$. Its exceptional part below8, if present, determines the outer branch $\ell$ uniquely. Polynomial convolution is exactly multiplication by $1+t+\cdots+t^r$, so induction on the factors proves every coefficient. Degrees never exceed57. All remaining operations are finite integer or rational sums and products.

The stated cardinality and minima are the computed proposition: the complete saved certificate records every rank tuple and coefficient list, and the independent endpoint checker reconstructs the domain and every convolution before comparing them. It evaluates all 2552 margins, rather than checking only degree, symmetry or a sample of rows. The two minima are separate assertions and need not have the same attaining profile. The implementation and records are identified in the reproducibility section. □

**Lemma 12** (Concentrating two simplex ranks)**.** *Let $1\le a\le b$ be integers, and let $U_r$ be uniform on $\{0,\dots,r\}$. With independent summands, $U_{a-1}+U_{b+1}$ dominates $U_a+U_b$ in convex order. Adding any common independent summand preserves this order.*

*Proof.* Both sums have mean $(a+b)/2$ and are symmetric about that mean. For an integer threshold $t$, let $\Delta_t$ be the difference of their expectations of $(X-t)_+$. Equal means allow us to use $(t-X)_+$ instead, and symmetry gives $\Delta_t=\Delta_{a+b-t}$. It suffices to take $0\le t\le(a+b)/2$. Put $D=(a+1)(b+1)$ and $D'=a(b+2)$. For $0\le t\le a$, direct summation of the increasing side of the two convolution triangles gives

$$\Delta_t=\binom{t+2}{3}\left(\frac{1}{D'}-\frac{1}{D}\right)\ge 0,\qquad D-D'=b-a+1>0.$$

For $a \le t \le (a+b)/2 \le b$, summing over the first uniform instead gives

$$\Delta_t = \frac{t(a+b-t) - (a-1)(a+2b+2)/3}{2(b+1)(b+2)}.$$

The numerator is increasing up to $(a+b)/2$. At $t = a$ it is $(a+2)(b-a+1)/3 > 0$. Outside the common support the difference is zero; between integer thresholds it is linear. Thus all stop-loss differences are nonnegative. On the integer support the values of a convex function are an affine function plus a nonnegative sum of these stop-loss functions, proving convex order. Conditioning on a common independent summand proves the last assertion. □

**Lemma 13** (Positive low8 reduction)**.** *Every actual ideal vertex has* $\kappa - C_{\mathrm{act}} \ge 36$.

*Proof.* For$1 \le a \le b \le 7$, replacing$(a, b)$ by$(a-1, b+1)$ preserves normalized envelope cost, since all these ranks stay at most eight. Lemma 12, including its independent-summand assertion, gives the required convex order. Since

$$\frac{G}{V} = \frac{57 - \mathbb{E}|2S - 57|}{2},$$

the transfer does not increase normalized curvature or normalized margin. Its product decreases from$(a+1)(b+1)$ to$a(b+2)$. Iteration terminates at ranks at least 8 and at most one positive rank below8. Indeed the integer potential $\sum_i r_i^2$ increases by $2(b-a+1) \ge 2$ at every step and is bounded above by $57^2$. Its initial value is at least 57, so at most 1596 steps occur. Rank zero is omitted. No transfer passes through the changed rank 9 envelope. The stopping domain is exactly (5).

Proposition 11 gives positive normalized terminal margin and absolute envelope surplus at least 36. If the original and terminal quantities are$(V, m)$ and$(V', m')$, then$V \ge V'$ and$m \ge m' > 0$. Thus $Vm \ge V'm' \ge 36$. This multiplication uses positivity. Finally (1) transfers the same lower bound to actual cusp charge. □

## 6 Shape curvature and the contradiction

**Lemma 14.** *For a proposed ambient 58 polytope,*

$$\mathcal{D}_5 \ge 377 v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

*Proof.* Choose a generic linear function on the bounded Klein closure of the polytope. Let $E_v$ be the number of edges of its simple vertex figure and $E_v^0$ the number not cut by the induced level section. If $I$ counts all polygon vertices with multiplicity, the published incidence identities give $I = \sum_v E_v$ and $2f_2 = \sum_v E_v^0$ [3, Theorem9]. Hence $\mathcal{D}_5 = \sum_v (5E_v^0/2 - E_v)$.

Orient each vertex figure by its induced generic linear function. Assign an uncut edge below the section to its upper endpoint and one above the section to its lower endpoint. A vertex of index $j$ receives $j$ or $57-j$ contributions, at least their minimum. The index distribution is its $h$-vector [3, Theorems2 and 6], so $E_v^0 \ge G_v$. At the distinct global minimum and maximum the whole vertex figure is uncut, and symmetry gives

$$E_v - G_v = \sum_{j > 57/2} h_j (2j - 57) \ge \sum_{j=29}^{57} (2j - 57) = 841.$$

Every coefficient of a simplex or product-of-simplices$h$-polynomial is at least 1. Adding the two extrema and multiplying by$5/2$ gives$4205$, counted once. For a finite vertex$h_j = 1$, so

$$G_0 = \sum_{j=0}^{57} \min(j, 57-j) = 812, \qquad \frac{5G_0 - 57 \cdot 58}{2} = 377.$$

The remaining ideal terms are precisely$\kappa$. □

For faces with at least five sides,$5 - s(F) \le 0$. The nonnegative local rule and Proposition 10 therefore give

$$\mathcal{D}_5 \le 377 v_f + \sum C_{\mathrm{act}}.$$

Subtract Lemma 14 and use Lemma 13:

$$0 \ge 4205 + \sum (\kappa - C_{\mathrm{act}}) \ge 4205 + 36 v_i > 0.$$

This also excludes the all-finite case. Thus dimension 58 is impossible. Appendices D and E exclude all dimensions at least$59$. This proves Theorem 1.

## 7 Reproducibility and assumption boundary

The final endpoint command is

```
python3 scripts/fresh_d13_context_dimension57_20261018.py --verify
```

It checks the frozen common bindings, the complete eight-obstruction and six-increasing-forest closure, all 2552 polynomial products and margins, all 28 stop-loss transfers and the constants$377$,$4205$. It does not repeat finite Bellman tables, native-support censuses or historical geometry. Those certificates have their own separate replay commands. The new local producer recomputes its$33$ contexts, exact inverse profiles, physical stars and marked-role interfaces; the written arbitrary-support argument remains part of its proof, not a finite absence test.

Published inputs are finite/affine Coxeter classification, the minimal indefinite order ceilings, standard finite-volume cusp/simple-edge facts, the incidence/index formulas, and Nikulin's narrow-facet theorem for the unbounded tail. The new local transfers, whole-factor seals, rank 58 guards, typed-context budgets, factor-fiber multiplicity, convex-order reduction, strict extrema correction and completed dimension tails are internal obligations. They may not be imported as external axioms merely because their proofs are lengthy.

The Lean project is separate and partial. Its checked generic and executable arithmetic results do not by themselves formalize all geometry or the complete dimension theorem. The companion lemma map specifies quantified internal interfaces rather than axioms for these new results. No successful hash or TeX build is represented as a proof check.

## A Geometric coverage and exact local certificates

The finite objects in this appendix are marked Gram diagrams and certificates for their extensions. They are not a finite list of all supporting forests. The extension arguments are what allow arbitrarily many additional supporting roots. We first describe the domains and checking predicates, then state the recorded finite propositions. The same organization applies to the preceding rule and to the final rule, with the final guarded consumer treated in the body of the paper.

## A.1 The complete core domain

Generate the connected affine diagrams of ranks four through eight: the four classical families in each rank, together with $\widetilde{F}_4, \widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8$. These are 24 sources. To each source on $\ell$ roots attach a root by every nonzero contact vector in $\{0, 1, \sqrt{2}\}^\ell$. Keep an extension exactly when it is indefinite and every one-root deletion is a finite or affine forest. Identify labelled isomorphic results, and retain every ordered pair of marks whose two deletions are positive definite. This is a finite exact algorithm; isomorphism testing may be done by trying all permutations of at most ten roots.

It is complete. A minimal indefinite core of order at least six cannot be a compact Lannér diagram. It has a proper affine block; the positive kernel and connectivity show that adjoining one adjacent root already makes an indefinite block. Minimality therefore makes this the whole core. A label at least five would occur in a connected proper five-root subdiagram, forbidden by finite/affine classification. A parallel pair with a third contacting root is already indefinite, and a dotted pair is indefinite alone. The contact alphabet is therefore a consequence of classification, rather than a cutoff. Checking all one-root deletions checks every proper principal block.

For orders four and five generate every connected graph with labels respectively 2 through 6 and 2 through 5, and apply the same acceptance and marking tests. These alphabets follow from the finite/affine three- and four-root lists. Order-two dotted cores are analytic cases. For an order-three core retain the full Gram family

$$\begin{pmatrix} 2 & -x & -u \\ -x & 2 & -v \\ -u & -v & 2 \end{pmatrix}, \qquad x = 2\cos(\pi/m),$$

including the parallel endpoint $x = 2$. The black–black label is unbounded; it is not enumerated up to a numerical maximum.

## A.2 All unions and all additional cross bonds

In a marked core $C$ put $\delta_C(v) = \min(d(v, a), d(v, c))$. The high-core catalogue has $\delta_C \le 7$. In the branch with no adjacent marked path of length at most eight, a shared white pair $(v, w)$ must satisfy $\delta_C(v) + \delta_D(w) \ge 9$. Generate every injective partial matching of white nodes satisfying this condition and preserving all labelled edges and nonedges. No black node is identified. This bounded matching loop includes failed identifications.

In the resulting union test all remaining permitted cross bonds and all permitted multi-position subsets. A cross bond at too small a combined distance gives an earlier short path. The inverse tests on the positive white blocks and actual black deletions give a finite native alphabet at every other position. For example a bond of magnitude $b$ between two positive white blocks requires

$$b^2 (G_{C,W}^{-1})_{vv} (G_{D,W}^{-1})_{ww} < 1.$$

Inverse positivity makes the corresponding necessary one-contact bounds valid even in the presence of further nonnegative contacts.

A pruning certificate must exhibit one of the following: a nonzero nonnegative vector of nonpositive square in a block required positive definite; a vector of strictly negative square in a corner required positive semidefinite; negative index at least two; or a retained adjacent marked short path. A nonnegative-vector witness at minimum native magnitudes remains valid when further contacts are added, since its square can only decrease. Negative index by itself is not used as a monotonicity assertion for further bonds; those branches are enumerated separately.

The unbounded order-three clause has an exact symbolic certificate. Its only surviving high port has inverse diagonal zero. Eliminating the high core gives zero Schur correction to the displayed small Gram for every physical $x, u, v$. Its negative direction then supplies a second negative direction. This disposes of every finite label and the parallel endpoint at once.

### A.3 What an extension certificate means

The common geometric lemmas in Section 2 give the following complete meanings to the certificate types. A radical certificate lists the possible contact positions and a nonzero full-core kernel vector. Since its represented vector is zero, every new contact annihilates it. If all allowed coefficients are nonzero, a nonempty nonnegative contact must meet both sign sets. The certificate lists the resulting short paths. Otherwise every extra root is orthogonal to the core, including the first root of any indirect extension.

An inverse certificate gives the physical core inverse $J$. A new norm-two root with contact column $q$ has residual square $2-q^tJq \geq 0$ in a positive-definite complement. At equality its residual vector is zero, so all further products are fixed. For two prospective contacts at $i, j$, compare $J_{ij}^2$ with $(2 - J_{ii})(2 - J_{jj})$; Cauchy either excludes them or forces the displayed bond and short path. Multiple-contact exclusions use nonnegative inverse entries in the appropriate end block.

An attached-forest certificate gives a single remaining old contact position $p$ with $J_{pp} = r > 1$. The argument of Lemma 8 bounds the entire new rank by $1/(r - 1)$, including indirect tails. A protected-role certificate instead forbids a first new contact at specified incident neighbors, thereby preserving a feature or a leaf of the *whole* component. An affine certificate supplies a positive kernel at an actual cusp; all further supporting contacts are orthogonal to that entire factor.

Each certificate is checked in the physical Gram normalization. Positive diagonal congruences may be used only with their norm scales recorded. An unbounded parameter is handled by a proved symbolic identity or a polynomial nonnegativity certificate on its whole stated range, never by sampling.

**Lemma 15** (A terminal affine-$B$ tail)**.** *Let $K$ be a nondegenerate Lorentzian subspace of the physical root space. Suppose that a root $e$ of squared norm 2 has orthogonal residual $e_\perp$ with $e_\perp^2 = \rho < 1$. There cannot be a chain $f_1, \dots, f_k$ of roots in $K^\perp$, each of squared norm 2, with $k \geq 1$ and the following native products. For $k \geq 2$, consecutive products are $-1$, except for $(f_{k-1}, f_k) = -\sqrt{2}$; also $(e, f_1) = -1$. For $k = 1$, instead $(e, f_1) = -\sqrt{2}$. All other products between distinct roots in the chain, and between $e$ and $f_j$ for $j > 1$, are zero.*

*Proof.* The complement $K^\perp$ is positive definite. Put

$$v = f_1 + \cdots + f_{k-1} + \frac{f_k}{\sqrt{2}}.$$

For $k \geq 2$, its diagonal terms sum to $2(k-1)+1$ and its consecutive cross terms sum to $-2(k-1)$. Thus $v^2 = 1$. For $k = 1$ the same identity follows from $f_1^2 = 2$. The prescribed products give $(e, v) = -1$ in both cases. Since $v \in K^\perp$, positive-space Cauchy gives the contradiction

$$1 = (e_\perp, v)^2 \leq e_\perp^2 v^2 = \rho < 1.$$

Only the coefficient of the final root has changed: every physical root still has squared norm 2. □

This lemma treats a terminal chain of any length, including cases where the first two new roots alone do not violate Cauchy. Its orthogonality hypothesis must be proved before it is used. After

excluding an adjacent white-interior path of length at most eight, the complete contact certificate permits a unique first root, or permits that root together with one exterior root. In the latter case the full no-bridge estimate bounds the whole component in the presence of the exterior root; that bound is strictly smaller than the actual target order. This excludes the exterior root and every indirect bridge. First-contact uniqueness then proves successively that all later roots are genuinely new and orthogonal to the old core. Their native zero products can be used only after this step. If the sole old-core contact of the first root is at $i$ with magnitude 1 and $J_{ii} > 1$, the inverse certificate gives $\rho = 2 - J_{ii} < 1$.

The exact application checks the complete supplied terminal-suffix map, all its edges and nonedges, and the original physical contact-profile selector. A contact theorem computed for another target is used only when its original core, ordered corner and full contact hypotheses coincide; its old target map is never relabelled as the present one. The terminal congruence in the calculation of $v^2$ is a rational check, with the scale $1/\sqrt{2}$ recorded explicitly.

For every bounded whole component, enumerate each classical target at every allowed rank or order, every exceptional target at its rank, and every induced labelled embedding of its displayed subdiagram. Assign target vertices successively and check each edge and nonedge against the earlier assignments. Finite target size proves termination; deleting the last assigned vertex proves completeness by induction. The marked images select individual entries of Appendix B, including zeros and fractions. The displayed whole-component bound, rather than an arbitrary search cap, proves that every actual completion is represented.

**Proposition 16** (Recorded core and extension calculations)**.** *The preceding bounded algorithms have the following exact outputs. The high-core generator tests 146 262 attachments and gives 26 isomorphism classes, with order counts* $12, 3, 4, 4, 3$ *in orders 6 through 10 and 768 ordered finite-deletion markings. The low order-four and order-five counts are 32 and 14. The complete high-overlap domain has 82 197 consistent matchings; 75 947 inverse and 3636 edge prunings leave 2614 contexts on 154 physical cores. They have respectively 1329 radical, 823 positive-residual, 259 bounded-component, 120 isolated-inverse, 74 protected and 9 adaptive certificates. All free cross bonds are discharged by 154 502 short-path and 5528 edge witnesses.*

*Every disjoint high/high union is excluded, including all 10 990 multi-position subsets. The low/high domain is excluded or gives a whole component of order at most twelve or an independent short path. Bounded affine target searches use all 1153 eligible patterns in orders 14–17, with whole-bound multiplicities* $313, 354, 346, 140$. *The exact extension certificates satisfy the predicates just specified. In each retained terminal configuration their chosen* current *corner contributions sum to at least one.*

This is a computed proposition, supported by the separate exact local certificates in the reproducibility package. Its assertion includes the receipt identities and inequalities, not merely the listed counts. The finite orders, alphabets, matchings and target bounds prove termination. The core, union and extension arguments prove coverage. Successful rows alone would prove neither.

## A.4 Triangles and their four exceptional markings

**Lemma 17** (The reduced small affine pairs are absent from triangles)**.** *No actual triangular two-face has an ideal corner whose varying affine factor and marked pair are one of the following:*

1. *a cycle of order ten, marked at antipodal nodes;*

2. *a cycle of order eleven, marked at distance five;*

*3. $\widetilde{D}_9$ of order ten, marked at leaves of opposite forks.*

*This assertion allows arbitrary additional supporting roots and other factors of the cusp.*

*Proof.* Let $a, b$ be the two marked roots at the ideal corner, let $F$ be its varying affine factor, and let $c$ be the third side root of the triangle. Normalize every root to square two. All nodes of $F \setminus \{a, b\}$ support the triangle. The other two actual corner diagrams are positive semidefinite, so both principal sets $(F \setminus \{a\}) + c$ and $(F \setminus \{b\}) + c$ are positive semidefinite.

For either positive-definite deleted-factor matrix $B$, put $p_j = -(c, r_j) \geq 0$. Schur complementation gives

$$p^T B^{-1} p \leq 2.$$

The inverse is entrywise nonnegative, and every nonzero native contact has magnitude at least one. Consequently a position with $(B^{-1})_{jj} > 2$ cannot be contacted. This is a necessary condition on a principal submatrix; extra supporting roots cannot repair its failure.

Number a cycle of order $m$ by $0, \ldots, m-1$, and take $a = 0, b = 5$. Deleting a node gives $A_{m-1}$, whose inverse diagonal at path position $k$ is $k(m-k)/m$. For $m = 10, 11$, all positions at least three steps from both path ends have diagonal at least $21/10, 24/11$, respectively, both greater than two. The two deleted-factor tests therefore give these possible contact sets:

| $m$ | after deleting 0 | after deleting 5 |
|---|---|---|
| 10 | $\{1, 2, 8, 9\}$ | $\{3, 4, 6, 7\}$ |
| 11 | $\{1, 2, 9, 10\}$ | $\{3, 4, 6, 7\}$. |

In each row the sets are disjoint. Contact to either removed mark is forbidden by the other deletion: the relevant diagonal is $5/2$ at order ten and $30/11$ at order eleven. Thus $c$ is orthogonal to every root of $F$.

For the last case number $F$ as the spine $0, \ldots, 5$, with leaves $6, 7$ at 0 and $8, 9$ at 5, and take $a = 6, b = 8$. Each deletion is finite $D_9$. After deleting 6, its only inverse diagonals at most two occur at the long endpoint 7 and its neighbor 0, with values one and two. After deleting 8, they occur at $9, 5$ with the same values. The short-leaf diagonals are $9/4$, and all other diagonals exceed two. Hence the possible contact sets are $\{0, 7\}$ and $\{5, 9\}$, which are disjoint. The other deletion also forbids contact to each removed mark, again by its short-leaf diagonal $9/4$. Thus $c$ is orthogonal to all of $F$ in this case as well.

Let $\nu = \sum_{j \in F} \delta_j r_j$, with every $\delta_j > 0$, be the positive affine-kernel combination. It is a nonzero future null vector representing the actual ideal corner: nonzeroness follows by pairing with a strict interior point. The third side of a nondegenerate actual triangle does not contain that corner, so $-(c, \nu) > 0$. The orthogonality just proved instead gives $(c, \nu) = 0$, a contradiction. Rotation and reflection cover all cycle pairs, and fork interchange and spine reversal cover all four opposite-fork pairs. The argument has used no restriction on spectator rank or on the other cusp factors. □

We now compare the inherited triangle witnesses with the actual weights of Appendix B. The two cycle orbits above are the only current reductions of ideal paths of length at most five. The reduced order-ten $D$ orbit has distance seven and is also absent from triangles by the lemma. Every other printed replacement in affine $B, C, D$ of orders ten through seventeen has distance at least eight. For larger orders the selected length-six or length-seven feature paths retain their units: their branch or nonsimple-bond witness persists within five steps of a mark. Such a witness cannot lie in an affine $A$ cycle. Every affine factor of order at most nine still pays a unit on every pair.

The four alternate witnesses below have an additional protection. Their fixed simple leaf with a degree-two neighbor excludes every classical affine factor, while exceptional affine factors have order

at most nine and retain full units. If the marked component is finite, the protected leaf is at distances $2, 1, 0, 0$ from a mark in the four displayed cases. It remains a whole-component endpoint, so the deep finite deletions, which require both marks to be at least six steps from every finite feature, do not remove the unit. The other selected finite length-six or length-seven witnesses already have a persistent branch or nonsimple feature within five steps. The sparse finite replacements in ranks twelve through seventeen are all longer than these selected paths. Thus the two-unit triangle argument below applies to the current rule, rather than assuming that every historical short ideal pair still has weight one.

A minimal indefinite core of an actual triangle contains all three black roots, since every deletion of one black lies in a positive semidefinite corner. Up to order six, a Steiner tree through the marks gives two distinct corner paths avoiding the third black, each of length at most five. In higher orders generate every marked triple in every core: there are $3\binom{7}{3} + 4\binom{8}{3} + 4\binom{9}{3} + 3\binom{10}{3} = 1025$. Delete the third black when evaluating each corner path and feature. The exact elementary test gives two persistent payments except at four triples in the simple ten-node tree

$$0{-}1,\ 0{-}2{-}3,\quad 0{-}4{-}5{-}6{-}7{-}8{-}9.$$

Their protected alternate pairs are

| Triple | Alternate pair | Length | Fixed leaf and neighbor |
|---|---|---|---|
| $(0, 1, 9)$ | $(0, 9)$ | 6 | $3, 2$ |
| $(0, 2, 9)$ | $(0, 9)$ | 6 | $1, 0$ |
| $(1, 2, 8)$ | $(1, 8)$ | 6 | $1, 0$ |
| $(1, 2, 9)$ | $(1, 9)$ | 7 | $1, 0$ |

In the first, second and fourth cases, the other corner contains affine $E_8$ on $0, \dots, 8$ with positive kernel $(6, 3, 4, 2, 5, 4, 3, 2, 1)$. Kernel orthogonality forbids all additional supporting contacts there. In the third case the finite $E_8$ inverse on $0, \dots, 7$ has diagonal $(30, 8, 14, 4, 20, 12, 6, 2)$ and is entrywise nonnegative. A new native contact at any of $0, \dots, 6$ has projection above two; only position 7 can be contacted. The displayed leaf and its neighbor therefore remain protected in every case.

No classical affine diagram has a simple leaf with a degree-two neighbor. Thus the alternate lies in a finite component or in an exceptional affine factor of order at most nine, and its unit persists. The complete elementary test also verifies the persistent feature for all 54 distance-six cases and the distance-seven cases. The deep finite deletions and the sparse long-pair changes affect none of these selected units. This proves triangle payment two.

## A.5 The order of the quadrilateral arguments

The local statements are composed by complete face inequalities, not by recursively replacing an obsolete corner unit. Write $S_k$ for the assertion that a suitable adjacent marked path of length at most $k$ makes the *whole face* receive at least one, and $F_k$ for the replacement of a deleted deep finite length-$k$ payment. The assertion $S_k$ does not give unit weight to that particular pair. Direct literal roles and the independent cycle-ten and cycle-eleven arguments first prove $S_5$. Their second halves occur at distinct corners, and their whole finite leaves and small-factor roles have the current printed weights. The independent near-terminal affine-$C$ zero proofs likewise pay the entire face; their zero corner is never called a unit.

One more independent priority is essential. An opposite-fork pair in the affine $D$ diagram of order ten contributes $27/32$, rather than its historical unit. Its complete four-profile determinant argument has at least one extreme end. With exactly one extreme, full-metric Cauchy either gives a retained short path or makes its entire spectator column empty or a single unit endpoint contact

on a whole $A_r$, $r \le 3$. With two extremes, the difference energy is $7/4$. A close mixed component gives a distinct short-five half. The complete far inverse spectrum leaves whole $A_{15}$ or $D_7$; their literal corners give a restored $A_{17}$ endpoint, a protected $D_{25}$ unit, or a whole exceptional affine unit. In the unmixed case choose a side of energy at most $7/8$, giving either an empty column or one unit endpoint contact on a whole $A_r$, $r \le 7$. In either unmixed branch put $r = 0$ for the empty column. If the extreme root's old unit attachment meets the selected leaf, the paying corner has a whole $A_{r+1}$ endpoint and a whole $D_9$ short leaf. The printed coupon is $5/32$. If instead it meets the old mate, the corner is whole $D_{10+r}$ with a marked short leaf at distance eight, a retained unit for every $0 \le r \le 7$. These are two distinct attachment cases, not two descriptions of the same coupon. The immediate affine alternatives also supply a distinct literal half or $27/32$. Thus every branch pays at least $27/32 + 5/32 = 1$, without recursively treating this corner as a unit.

Here the independent computed inputs are the complete far inverse blocks and their strict-gradient inequalities, all sixteen whole finite coupons, and all twelve mixed whole-corner Gram records in `scripts/fresh_order10_d_opposite_fork_fractional_transfer_20261018.py` and its exact saved certificate. Their finite domain is explicit: for an $A$ component, unit difference energy $d(N-d)/N \le 7/4$ and strict separate inverse diagonals imply

$$4 \le d \le 14, \qquad d + 2 \le N \le \left\lfloor \frac{4d^2}{4d-7} \right\rfloor .$$

Enumerate every position in those ranges satisfying the strict diagonal tests. Complete finite inverse formulas leave only $D_6, D_7$ and the eligible $E_6$ block outside $A$; other types have no far allowable pair. Nonunit quadrants have their minimum at $(\sqrt{2}, 1)$ or $(1, \sqrt{2})$, by the recorded positive gradients and positive quadratic remainder. Their minima exceed $7/4$. All loops are finite and the inverse bounds prove coverage; the literal Gram records check every actual corner and its whole marked role. These inputs are not inferred from the counts in Proposition 16.

The remaining dependency order is

$$S_5 \longrightarrow F_6 \longrightarrow S_6 \longrightarrow F_7 \longrightarrow S_7 \longrightarrow F_8 \longrightarrow S_8.$$

Here $S_6$ uses the cycle cases with shorter arc six, including the order-fourteen case, together with $F_6$. The protected affine-$D$ order-ten terminal of $F_7$ invokes the independent priority just proved. Before using $S_7$, prove the cycle arcs $(7, 8)$ rectangle bound separately: a required neighboring contact gives a Cauchy correction on one fixed side of the entire rectangle. The corrected quadratic remains concave in reciprocal heights. This calculation has no $F_8$ premise, even though its certificate shares a file with that later composition. For arcs $(7, 7)$ the sealed equality corners currently pay $41/64 + 23/64 = 1$, rather than two historical halves. The affine-tree cases use their persistent branch or nonsimple-bond features and the independent small-factor priorities. Finally the strict $F_8$ corridor ends at $S_7$, after which the remaining cycle and tree cases prove $S_8$. All additional supporting contacts are controlled by the whole-component seals, not by an ambient-rank cap. The large finite-$D$ leaf-nine half has its own distinct small/$D_{10}$ half; it is not used as a unit.

For a small affine factor, its positive kernel first splits off the spectator forest. Adjacent black contact or a shared spectator neighbor gives an earlier short path. Otherwise the two spectator columns have disjoint support. The complete inverse-diagonal profiles and four-root Schur determinant give a strict obstruction or a closed equality system. At equality Gram Cauchy determines the active whole spectator types. These are the bounded profile calculations in Proposition 16, with unbounded native magnitudes retained symbolically. A changed affine-$B$ corner invokes its independent full-face theorem before a later affine-$D$ theorem can spend that corner's old half. An immediate affine alternative within one equality receipt contributes its current fraction at a distinct corner and makes no recursive theorem call.

Finally the finite sparse substitutions are single inequalities on all four actual corners. Their relevant contribution lists are $(1/2, 1/4, 1/4, 0)$, $(0, 1/8, 1/4, 5/8)$ and $(0, 1/4, 1/4, 1/2)$, each summing to one. The extension seals prove the whole context witnesses at the positive corners. If a witness is absent, that branch is excluded or independently paid; an induced small subgraph cannot serve as a witness. The full chosen-role checks cover the 216 literal stars, 564 later corners, 24 $D_{14}$ variants and 332 positive $D_{12}$ outputs. Universal outputs outside those fixed Gram lists use their whole-factor profile proofs. In particular, the affine-$B$ singleton consumers of $B_{13}(0, 12)$ have a distinct sealed $A_1/D_{12}$ unit, and the quiet affine-$D_{12}$ consumer of the guarded $D_{13}$ half is proved in the body of the paper.

This order is well founded: independent face payment, shorter-path or protected terminal payment, whole-factor profile system, then joint finite receipt. A joint receipt is one terminal inequality, not a call back to another decreased role. Every actual quadrilateral either ends at an earlier face theorem or reaches a union and extension certificate in the complete generated domain. Its distinct current contributions total at least one. This proves the local coverage used in Proposition 6 and, with Theorem 7, the final rule. It uses the recorded distributed certificates; no new flattened certificate run is assumed or claimed.

# B The intrinsic corner weights

We specify a single function of the actual marked vertex diagram. At a finite vertex let $\mathcal{F}$ be the full spherical forest, with two marked nodes $a, b$. In one whole component use the same-component entry below. In distinct components use the ordinary cross entry and the stated contextual additions. At a cusp use only the affine table for its varying factor, with the multiplicity proved in Section 2. An omitted role has weight zero.

Number $A_r$ by $0, \dots, r-1$, and $B_r$ by the same path with its last bond doubled. Number $D_r$ by the spine $0, \dots, r-3$ with leaves $L_r = \{r-2, r-1\}$ at the center $r-3$; its long end is 0. For an $A_r$ factor put $E_r = \{0, r-1\}$ and $P_k(A_r) = \{k-1, r-k\}$, removing repetitions. The long ends of $E_6$ lie on its two length-two arms. The tables are invariant under diagram automorphisms.

## B.1 Finite same-component entries

Every pair in a component of rank at most 11 has weight one; this includes $I_2(m)$ for every $m \geq 5$. For classical rank $r \geq 18$, a feature is a vertex of degree other than two, or an endpoint of a nonsimple bond. Put $\delta(v) = \operatorname{dist}(v, \text{features})$. Start with one at distance at most eight and at distance nine containing a whole-component leaf. Delete the unit at distance 6, 7 or 8 when both marks have $\delta \geq 6$. In $D_r$, $r \geq 20$, halve each distance-nine pair containing a short fork leaf. These are all ordinary large-rank entries. Their sums, with $x_+ = \max(x, 0)$, are

$$
\begin{aligned}
B(A_r) &= 8r - 34 - (r-18)_+ - (r-19)_+ - (r-20)_+, \\
B(B_r) &= 8r - 34 - (r-19)_+ - (r-20)_+ - (r-21)_+, \\
B(D_r) &= 8r - 32 - [r \geq 20] - (r-20)_+ - (r-21)_+ - (r-22)_+.
\end{aligned}
$$

For ranks 12–17 start with all units and make exactly the following replacements. Z, H, Q, O mean $0, 1/2, 1/4, 1/8$; A13 means $A_{13}$, and similarly for the other labels. Unlisted entries retain one. The $B_{17}$ convention is stated separately.

```
D12 H: (0,10),(0,11).
```

```
A13 Z: (0,12).
B13 Z: (0,12).
D13: no ordinary replacement.

A14 Z: (0,12),(1,13),(0,13).
B14 Z: (0,13),(1,13),(2,13),(3,13).
D14 Z: (0,12),(0,13); H: (0,11).

A15 Z: (0,12),(0,14),(2,14),(0,13),(1,14).
B15 Z: (0,12),(0,13),(0,14),(1,13),(1,14),(2,14),(3,14),(4,14).
D15 Z: (0,11),(1,13),(1,14);
    H: (0,12),(0,13),(0,14),(1,12).

A16 Z: (0,12),(0,14),(0,15),(1,14),(1,15),(3,15),(1,12),(3,14);
    H: (0,13),(1,13),(2,14),(2,15);
    O: (0,11),(4,15).
B16 Z: (0,11),(0,12),(0,13),(0,14),(0,15),
       (1,12),(1,13),(1,14),(1,15),(2,13),(2,14),(2,15),
       (3,14),(3,15),(4,15),(5,15).
D16 Z: (0,10),(0,11),(0,12),(0,13),(1,11),(2,12),(2,14),(2,15);
    H: (0,14),(0,15),(1,12),(1,13),(1,14),(1,15),(2,13).

A17 Z: (0,10),(0,11),(0,12),(0,14),(0,15),
       (1,10),(1,11),(1,13),(1,14),(1,16),
       (2,12),(2,14),(2,15),(2,16),(3,15),(4,14),(4,16),
       (5,15),(5,16),(6,15),(6,16);
    H: (0,16),(2,13),(3,14);
    weight7/8: (1,12),(4,15).
D17 Z: (0,10),(0,11),(0,12),(0,13),
       (1,10),(1,11),(1,12),(1,13),(1,14),(1,15),(1,16),
       (2,12),(2,14),(3,15),(3,16),(4,14),(4,15),(4,16);
    H: (0,14),(2,13),(2,15),(2,16),(3,13),(3,14);
    Q: (0,15),(0,16).
```

For $B_{17}$ keep one exactly at distance at most eight, at distance nine containing a whole-component leaf, and at $(2,11),(3,12),(4,13)$. All other entries are zero. The ordinary same-component sums are

| $r$ | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|
| $A_r$ | 66 | 77 | 88 | 100 | 433/4 | 453/4 |
| $B_r$ | 66 | 77 | 87 | 97 | 104 | 105 |
| $D_r$ | 65 | 78 | 177/2 | 100 | 217/2 | 227/2 |

For Proposition 6 alone restore $B_{13}(0,12) = 1$, so that its preliminary sum is 78. The final rule uses the zero entry above. This is the only ordinary-table difference in that preliminary calculation.

## B.2 Ordinary cross-component roles

The table is symmetric in the two marked components. A product of mark sets means every pair in that product. In the first seven rows $S$ is any irreducible component of rank at most five, and all($S$) is its node set. Apply those broad rows first, then replace values where a later row applies. Later rows are replacements, not additions; overlaps never disagree. All unlisted roles are zero. An arbitrary $I_2(m)$ is a small rank-two type, but is not an $A_2$ where that type is expressly named.

| First component and marks | Second component and marks | Weight |
|---|---|---|
| S,all($S$) | $D_{10}$,$L_{10}$ | 1/2 |
| S,all($S$) | $D_{11}$,$D_{12}$,$D_{13}$,$D_{14}$, center and short leaves | 1/2 |
| S,all($S$),rank($S$) $\leq 2$ | $D_{15}$,center and short leaves | 1/2 |
| S,all($S$),$3 \leq$ rank($S$) $\leq 5$ | $D_{15}$,$L_{15}$ | 1/2 |
| $A_1$ | $D_{16}$,center and short leaves | 1/2 |
| S,all($S$) | $A_{13}$ central node 6 or $A_{15}$ central node 7 | 1/2 |
| S,all($S$) | $A_{17}$,marks5,11 | 1/2 |
| $A_1$ or $A_2$,all marks | $D_{11}$,$L_{11}$ | 1 |
| $A_4$,$E_4$ | $D_{11}$,$L_{11}$ | 1 |
| $A_5$,$P_2$ | $D_{11}$,$L_{11}$ | 1 |
| $E_6$,long ends | $D_{11}$,$L_{11}$ | 1 |
| $D_5$,long endpoint0 | $D_{11}$,$L_{11}$ | 1 |
| $A_1$ | $D_{12}$,$L_{12}$ | 1 |
| $D_5$,$L_5$ | $D_{12}$,center and short leaves | 1 |
| $A_2$,all marks | $D_{13}$,$L_{13}$ | 1 |
| $A_2$,all marks | $D_{13}$,center10 | 23/64 |
| $A_1$ or $A_4$,$E_4$ | $D_{13}$,mark9 | 1/2 |
| $A_1$ or $A_2$,all marks | $D_{14}$,mark10 | 1/2 |
| $A_1$ | $D_{14}$,mark9 | 1/16 |
| $A_1$ | $D_{15}$,mark11 | 1/16 |
| $A_2$,all marks, or $A_3$,central node 1 | $D_{10}$,center7 | 1 |
| $A_1$ or $A_2$,all marks, or $A_3$,$E_3$ or $A_4$,$E_4$ | $B_9$,doubled terminal8 | 1/2 |
| $A_1$ | $B_{10}$,doubled terminal9 | 1 |
| $A_2$,all marks | $B_{10}$,doubled terminal9 | 1/2 |
| $A_1$ | $B_{11}$,mark9 | 1/2 |
| $A_1$ | $B_{12}$,mark9 | 1/2 |
| $A_r, E_r$, $1 \leq r \leq 8$ | $D_9$,$L_9$ | 5/32 |
| $A_r, E_r$, $1 \leq r \leq 4$ | $A_9$,central node 4 | 1/2 |
| $A_1$ | $A_{11}$,central node 5 | 1 |
| $A_2$,all marks | $A_{11}$,central node 5 | 1/2 |
| $A_1$ or $A_2$,all marks, or $A_4$,$E_4$ | $A_{11}$,$E_{11}$ | 1/2 |
| $E_6$,long ends, or $D_5$,long end0 | $A_{11}$,$P_2$ | 1 |
| $A_1$ | $A_{12}$,marks5,6 | 1 |
| $A_1$ | $A_{12}$,$E_{12}$ | 1/4 |
| $A_4$,$E_4$ | $A_{12}$,$E_{12}$ or $A_{12}$,$P_3$ | 1 |
| $D_5$,long end0 | $A_{12}$,$P_2$ | 1 |
| $D_5$,long end0 | $A_{12}$,$E_{12}$ | 3/8 |
| $D_5$,$L_5$ | $A_{12}$,$P_3$ | 1 |
| $A_1$ | $A_{13}$,central node 6 | 41/64 |
| $A_2$,all marks | $A_{13}$,central node 6 | 23/64 |
| $A_1$ | $A_{13}$,$E_{13}$ | 1/2 |
| $A_1$ | $A_{13}$,$P_2$ | 1/4 |
| $A_2$,all marks | $A_{13}$,$E_{13}$ | 1 |
| $A_4$,$E_4$ | $A_{13}$,$P_2$ | 1 |
| $A_4$,internal marks1,2 | $A_{13}$,$P_2$ | 1/2 |
| $A_2$,all marks | $A_{14}$,$P_2$ or $A_{14}$,$P_3$ | 1/2 |
| $A_1$ | $A_{14}$,$P_3$ or $A_{14}$,$P_4$ | 1/2 |
| $A_1$ | $A_{15}$,$P_3$ | 1/2 |
| $A_1$ | $A_{16}$,$P_4$ | 1/2 |
| $A_1$ | $A_{16}$,$E_{16}$ | 1/4 |

For example, the $D_5$ long-end/$D_{12}$ center entry is one half; the upgrade to one concerns short

$D_5$ leaves. There is no ordinary $A_1/B_8$ payment.

### B.3 Contextual entries

Every witness is a distinct unmarked whole component at the actual corner. Presence is Boolean. The first table replaces ordinary same-component values; the second adds to cross roles whose ordinary values are zero.

| Marked component and pair | Presence guard | Value |
|---|---|---|
| $A_{13}$,(0,12) | $A_1$ present | 1 |
| $A_{14}$,(0,13) | $A_1$ present | 1/2 |
| $A_{15}$,(0,13),(1,14) | $A_1$ present | 1/4 |
| $D_{15}$,(1,13),(1,14) | $A_1$ present | 1/4 |
| $A_{16}$,(1,12),(3,14) | $A_1$ present | 5/8 |
| $D_{16}$,(0,14),(0,15) | $A_1$ present | 1 |
| $D_{14}$,(0,12),(0,13) | $A_1$ or $A_2$ or $A_3$ present | 1/2 |
| $D_{13}$,(0,11),(0,12) | at least three $A_{15}/D_{15}$ components | 1/2 |
| $D_{28}$,(0,12) | at least two $A_{15}/D_{15}$ components | ordinary +1/2 |

| Marked pair | Presence guard | Addition |
|---|---|---|
| $A_4$ endpoint/$A_{12}$ second node | $A_1$ present | 1/8 |
| $A_2$ mark/$A_4$ endpoint | $A_1$ and $A_{10}$ present | 1/4 |
| $A_2$ mark/$D_5$ long endpoint | $A_1$ and $A_9$ present | 1/2 |
| $A_1/D_5$ long endpoint | $A_{11}$ present OR both $A_9$,$A_2$ present | 3/8 |
| $A_1/D_5$ long endpoint | $A_9$ and $A_2$ present | 5/8 |
| two distinct $A_2$ components,any marks | $A_{10}$ and at least two $A_1$ present | 9/32 |
| $A_{12}$ endpoint/$D_{16}$ long endpoint | at least two $A_{15}/D_{15}$ present | 1/2 |

The two $A_1/D_5$ additions deliberately add to one under the joint $A_9, A_2$ guard; $A_{11}$ does not repeat the 3/8 term. The $A_2/A_2$ marks lie in distinct components.

These tables define $b(T)$ and $j(T,U)$ in (2): $b(T) = 64B(T)$ sums ordinary same-component entries, and $j(T,U)$ is 64 times the sum of ordinary cross entries on two distinct components. For the preliminary calculation make only the stated $B_{13}$ restoration and omit the three rank-15-guarded changes. Counting the other contextual pairs gives exactly $\Psi$ in Appendix C; for example the $A_2/A_2$ row costs $64 \cdot 4(9/32)\binom{a_2}{2} = 36a_2(a_2-1)$. The final additional context cost is

$$-64d_{13}[a_{15}+d_{15} \geq 3] + (32d_{28} + 64a_{12}d_{16})[a_{15}+d_{15} \geq 2].$$

This identifies one function feeding both geometry and maximization.

### B.4 Affine factors

Let $m$ be the order, so the affine rank is $m-1$. Number $\widetilde{A}$ as the cycle $0, \ldots, m-1$; $\widetilde{B}$ as spine $0, \ldots, m-3$, double bond $(m-4, m-3)$, and leaves $m-2, m-1$ at 0; $\widetilde{C}$ as the path $0, \ldots, m-1$ with both end bonds doubled; and $\widetilde{D}$ as spine $0, \ldots, m-5$ with leaves $m-4, m-3$ at 0 and $m-2, m-1$ at $m-5$. In the following table A, B, C, D mean these affine families.

Every pair in order at most nine has weight one, including every exceptional affine type. For $\widetilde{A}$ at order ten halve its five antipodal pairs; at order eleven halve its eleven distance-five pairs; at every order at least twelve retain exactly distance at most five. For B, C, D at orders 10–17 start with all units and make these replacements. XxY is a Cartesian product.

```
m10:
 B H:(7,8),(7,9).
 C Z:(0,9).
 D weight27/32: {6,7}x{8,9}.

m11:
 B Z:(8,9),(8,10).
 C Z:(0,9),(0,10),(1,10).
 D H:{7,8}x{9,10}.

m12:
 B Z:(0,9),(9,10),(9,11); H:(8,10),(8,11).
 C Z:(0,10),(0,11),(1,10),(1,11).
 D Z:{8,9}x{10,11}.

m13:
 B Z:(0,10),(9,11),(9,12),(10,11),(10,12);
   H:(1,10),(8,11),(8,12).
 C Z:(0,10),(0,11),(0,12),(1,11),(1,12),(2,10),(2,12).
 D Z:(0,8),{9,10}x{11,12};
   H:(0,11),(0,12),(8,9),(8,10).

m14:
 B Z:(0,10),(0,11),(1,10),(1,11),(2,11),
     (9,12),(9,13),(10,12),(10,13),(11,12),(11,13).
 C Z:(0,10),(0,11),(0,12),(0,13),(1,11),(1,12),(1,13),
     (2,11),(2,12),(2,13),(3,13).
 D Z:(0,9),(0,12),(0,13),(9,10),(9,11),{10,11}x{12,13};
   H:(1,12),(1,13),(8,10),(8,11).

m15:
 B Z:(0,11),(0,12),(1,11),(1,12),(2,12),(3,12),
     {9,10,11,12}x{13,14}.
 C Z:all path pairs of distance>=10 except (2,12).
 D Z:(0,10),(0,13),(0,14),(1,13),(1,14),
     (9,11),(9,12),(10,11),(10,12),{11,12}x{13,14};
   weight15/16:(2,13),(2,14),(8,11),(8,12).

m16:
 B Z:all pairs of graph distance>=10.
 C Z:all path pairs of distance>=10 except (1,11),(4,14).
 D Z:(0,11),(0,14),(0,15),(2,14),(2,15),(9,12),(9,13),
     (11,12),(11,13),{12,13}x{14,15};
   H:(0,9),(0,10),(1,11),(2,11);
   weight7/16:(1,14),(1,15),(10,12),(10,13).

m17:
 B Z:all pairs of graph distance>=10.
 C Z:all path pairs of distance>=10.
```

```
D Z:(0,10),(0,12),(0,15),(0,16),(1,11),(2,12),
    (3,15),(3,16),(9,13),(9,14),(12,13),(12,14),
    {13,14}x{15,16};
  H:(0,11),(1,12),(1,15),(1,16),(11,13),(11,14);
  weight1/8:(2,15),(2,16),(10,13),(10,14).
```

For classical affine order at least 18, features are branching vertices and endpoints of nonsimple bonds. A simple endpoint alone is not an affine feature. With $\delta$ the distance to this set, keep a unit exactly when

$$d(a,b) \le 5 \quad \text{or} \quad 6 \le d(a,b) \le 9 \ \text{ and } \ \min(\delta(a),\delta(b)) \le 5.$$

All other entries are zero. A featureless cycle retains only distance at most five. There are no ideal context bonuses. Summing each table, taking the family maximum and dividing by order gives exactly $c(r)$ in Section 5; the large-rank counts give its formula for $r \ge 24$. Its geometric use is the factor-fiber argument establishing (1).

# C Contexts in the finite-forest calculation

This appendix specifies the finite algorithm behind Proposition 6. It concerns the rule before the last $B_{13}$ and three-rank-15 changes. All charges are multiplied by 64. Put $a_i = m_{A_i}$ and $d_i = m_{D_i}$, and write $[E]$ for the indicator of a condition $E$. The complete additional context cost in (2) is

$$\begin{aligned}\Psi(m) = {} & 36a_2(a_2-1)[a_1 \ge 2,\ a_{10} > 0] + 64a_2d_5[a_1 > 0,\ a_9 > 0] \\ & + 24a_1d_5[a_{11} > 0 \text{ or } (a_9 > 0,\ a_2 > 0)] + 40a_1d_5[a_9 > 0,\ a_2 > 0] \\ & + 32(a_{15}+d_{15})[a_1 > 0] + 64d_{14}[a_1+a_2+a_3 > 0] \\ & + [a_1 > 0]\big(80a_{16} + 32a_{14} + 64d_{16} + 64a_{13} \\ & \qquad + 32a_4a_{12} + 64a_2a_4[a_{10} > 0]\big).\end{aligned}$$

Witness presence is Boolean. Additional witnesses multiply a bonus only where an actual multiplicity occurs in this formula.

## C.1 Representatives and actual fibers

Use $E_6$ as the rank-six representative and $B_9$ as the rank-nine representative. At rank 12 identify $A_{12}, B_{12}$ with $A_{12}$, but keep $D_{12}$ separate. Keep $A_{17}$ separate from the other rank-17 types. At all remaining ranks at most eight use the type $A$ of that rank; at larger ranks use the type $D$ of that rank. The fixed representatives are generated by the integer domain

$$c_9, \dots, c_{17} \ge 0, \quad \sum_{r=9}^{17} r c_r \le 58, \quad 0 \le k_{12} \le c_{12}, \quad 0 \le k_5 \le \left\lfloor \frac{58 - \sum_{r=9}^{17} r c_r}{5} \right\rfloor.$$

They consist of $c_9$ copies of $B_9$, $c_{10}$ copies of $D_{10}$, $c_{11}$ copies of $D_{11}$, $k_{12}$ copies of $A_{12}$, $c_{12} - k_{12}$ copies of $D_{12}$, $c_r$ copies of $D_r$ for $13 \le r \le 16$, $c_{17}$ copies of $A_{17}$, and $k_5$ copies of $A_5$. The split $k_{12}$ counts representative $A_{12}$ components: before actual-type expansion it combines actual $A_{12}$ and $B_{12}$, and is distinct from the actual multiplicity $a_{12}$ in $\Psi$. The residual ranks are

$$\{1,2,3,4,6,7,8\} \ \cup \ \{17, \dots, 58\}.$$

Thus residual rank 17 represents $B_{17}, D_{17}$, whereas the $A_{17}$ count is already fixed. These conventions matter for the context bonuses.

## C.2 The numerical coarse screen

The coarse coefficients differ from the actual coefficients $b, j$: they include allowances for some presence guards. We define them here, so the preliminary screen is a finite numerical problem as well as a description of its output. Barred quantities are unscaled payments; multiplication by 64 gives integer coefficients. For the fixed multiset $F$ and remaining rank $R$ just defined, put

$$\mathcal{R} = \{1,2,3,4,6,7,8\} \cup \{17,\dots,58\}, \qquad u_r = \begin{cases} E_6 & r = 6, \\ A_r & r \le 8,\ r \ne 6, \\ D_r & r \ge 17. \end{cases}$$

The fixed base coefficients are

| $T$ | $A_5$ | $B_9$ | $D_{10}$ | $D_{11}$ | $A_{12}$ | $D_{12}$ | $D_{13}$ | $D_{14}$ | $D_{15}$ | $D_{16}$ | $A_{17}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\bar{b}(T)$ | 10 | 36 | 45 | 55 | 66 | 65 | 78 | 179/2 | 201/2 | 219/2 | 3613/32 |

For residual types, $\bar{b}(u_r) = \binom{r}{2}$ for $r \in \mathcal{R}$, $r \le 8$, while

$$\begin{aligned} \bar{b}(D_{17}) &= 919/8, \\ \bar{b}(D_r) &= 8r - 32 - [r \ge 20] - (r-20)_+ - (r-21)_+ - (r-22)_+ \qquad (r \ge 18), \end{aligned}$$

where $x_+ = \max(x, 0)$. The symmetric interaction table is as follows. Omitted entries are zero, including every interaction of two large fixed types and every interaction involving a residual $D_r$, $r \ge 17$.

| | $A_5$ | $B_9$ | $D_{10}$ | $D_{11}$ | $A_{12}$ | $D_{12}$ | $D_{13}$ | $D_{14}$ | $D_{15}$ | $D_{16}$ | $A_{17}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $A_1$ | 3/8 | 1/2 | 1 | 5/2 | 5/2 | 5/2 | 137/64 | 33/16 | 25/16 | 3/2 | 1 |
| $A_2$ | 0 | 1 | 4 | 5 | 0 | 3 | 151/32 | 4 | 3 | 0 | 2 |
| $A_3$ | 0 | 1 | 4 | 9/2 | 0 | 9/2 | 9/2 | 9/2 | 3 | 0 | 3 |
| $A_4$ | 0 | 1 | 4 | 8 | 17/2 | 6 | 8 | 6 | 4 | 0 | 4 |
| $A_5$ | 0 | 5/8 | 5 | 19/2 | 27/4 | 21/2 | 15/2 | 15/2 | 5 | 0 | 5 |
| $E_6$ | 0 | 5/8 | 0 | 4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $A_7$ | 0 | 5/8 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $A_8$ | 0 | 5/8 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Thus the $A_1/A_5$ entry is an unconditional upper allowance, not an ordinary payment. Set

$$K(F) = \sum_{T \in F} \bar{b}(T) + \sum_{\{T,U\} \subset F} \bar{j}(T,U), \qquad w_F(r) = \bar{b}(u_r) + \sum_{T \in F} \bar{j}(u_r, T).$$

Multiset sums count distinct occurrences, including equal types. In particular $K(F)$ includes all interactions of the fixed $A_5$ copies with the large fixed components. Residual representatives have no mutual interaction. Let $V_F$ be the exact-rank Bellman table with items $(r, w_F(r))$, $r \in \mathcal{R}$, and let $V_F^{\neg 2}$ omit rank two. Both are instances of Proposition 5, after scaling by 64.

There remain two context terms. For $s \in \{1,2\}$ set

$$G_0 = [s = 2,\ c_{10} > 0], \qquad G_1 = [c_9 k_5 > 0].$$

An admissible reservation has $s \le R$, $G_0 + G_1 > 0$, and $s = 2$ only if $G_0 = 1$. For every $0 \le j \le \lfloor (R-s)/2 \rfloor$ form

$$L_{s,j} = s w_F(1) + j w_F(2) + V_F^{\neg 2}(R - s - 2j) + \frac{9}{16} j(j-1) G_0 + j k_5 G_1.$$

Here $j$ counts all rank-two representatives: every actual rank-two type, including $B_2$ and the common dihedral symbol, maps to $u_2 = A_2$. It can exceed the actual $A_2$ count. Ignoring unreachable states, the coarse value is

$$U(F) = K(F) + \frac{11}{32}c_{17} + \max\bigl\{V_F(R),\ L_{s,j} : (s,j) \text{ admissible}\bigr\}. \tag{6}$$

The $11/32 = 22/64$ increment per fixed $A_{17}$ is separate from $\bar{b}(A_{17})$. These formulas define the recorded 3298-way screen and its 39 classes with $U(F) > 377$.

Here is the complete domination test behind this upper problem. Write $B(T) = b(T)/64$, $J(T,U) = j(T,U)/64$ for the preliminary actual rule, and let $\rho$ be the representative map. Put

$$g(A_{13}) = g(D_{14}) = g(D_{16}) = 1, \quad g(A_{14}) = g(A_{15}) = g(D_{15}) = \tfrac{1}{2}, \quad g(A_{16}) = \tfrac{5}{4},$$

and $g = 0$ otherwise. These are the additional same-component allowances from the presence guards, rather than their full replacement values. Let $h$ be symmetric, with $h(A_1, D_5) = 3/8$, $h(A_4, A_{12}) = 1/2$, and all other entries zero. The finite comparison checks, for every actual type and every actual type pair,

$$\begin{aligned} B(T) + g(T) &\le \bar{b}(\rho T) + \tfrac{11}{32}[T = A_{17}], \\ J(T,U) + h(T,U) &\le \bar{j}(\rho T, \rho U). \end{aligned}$$

This states the full domain and inequalities of the coefficient check; it is not a comparison only of the eventual maximizing forests.

Outside the two separately bounded families below, the remaining contexts are exactly the $A_2/A_2$ and $A_2/D_5$ terms in $L_{s,j}$. If the first occurs, reserve two $A_1$ and choose $s = 2$; this also includes the second term whenever it occurs. If only the second occurs, reserve one $A_1$ and choose $s = 1$. Choose $j$ to be the complete rank-two representative count. Actual $A_2$ counts are at most $j$, actual $D_5$ counts are at most $k_5$, and actual $A_9, A_{10}$ presence implies the corresponding representative presence. Monotonicity of $j(j-1)$ and $jk_5$ on nonnegative integers proves domination. If neither context occurs, $V_F(R)$ suffices. Further $A_1$ copies remain available in the residual table; they do not multiply a Boolean witness bonus. The reservations jointly cover every actual forest in this complement; an individual branch need not bound another formal reconstruction. The additional $5/8$ on $A_1/D_5$ under the joint $A_9, A_2$ guard and the $A_1, A_2, A_4, A_{10}$ family are covered by their separate complete conditional calculations, including overlap. No actual attainment of the coarse upper bound is asserted.

The ordinary component and interaction inequalities give an upper relaxation for every representative choice. The table check uses every actual type, including the common-cost dihedral symbol; it is not a comparison of only the eventual maximizers. It retains the separate increment of 22 per fixed $A_{17}$. There are 1295 fixed large vectors, 3298 choices after adjoining the rank-five count, and two residual tables per choice, one allowing $A_2$ and one excluding it. Every such table is checked by Proposition 5.

### C.3 The actual fibers of the remaining classes

Exactly 39 coarse classes can exceed 377, outside the two context families below. Expand *every* component of each such class into its full actual-type fiber. For repeated representatives choose a nondecreasing list of actual types. This enumerates all 748 fixed actual multisets $F$ without losing mixed assignments. Let $R = 58 - \operatorname{rank}(F)$ and enumerate the complete actual counts

$$0 \le a_1 \le R, \qquad 0 \le a_2 \le \left\lfloor \frac{R - a_1}{2} \right\rfloor.$$

Remove both $A_1$ and $A_2$ from the residual alphabet. If $a_1, a_2 > 0$ and $F$ contains $A_{10}$, also exclude residual $A_4$: those forests belong to the second family below.

At these fixed counts the only remaining presence condition is the $D_{14}$ bonus triggered by $A_3$, when $a_1 = a_2 = 0$ and $F$ contains $D_{14}$. Split this case into no $A_3$, or at least one $A_3$. In the second branch reserve one $A_3$ and charge the bonus $64m_{D_{14}}$ once; further $A_3$ copies remain ordinary residual items. In the first branch exclude $A_3$ entirely.

For an allowed residual type $T$, set

$$w_F(T) = b(T) + \sum_{U \in F} j(T, U).$$

Every ordinary interaction between residual types is zero, as checked on the whole residual alphabet. If $a_1 > 0$, a residual $A_4$ additionally receives $32m_{A_{12}}$. All other terms of $\Psi$ have become fixed. With $\epsilon \in \{0, 1\}$ denoting the reserved-$A_3$ branch, evaluate

$$\begin{aligned} C_{64}^{\mathrm{ord}}(F) + a_1 w_F(A_1) + a_2 w_F(A_2) + \Psi(F + A_1^{a_1} + A_2^{a_2}) \\ + \epsilon\big(w_F(A_3) + 64m_{D_{14}}\big) + V(R - a_1 - 2a_2 - 3\epsilon). \end{aligned}$$

Here $C_{64}^{\mathrm{ord}}$ omits $\Psi$, and $V$ uses the adjusted residual weights. Ignore unreachable final ranks. The two $A_3$ branches are used only in the specified case, so the displayed $\Psi$ has not already charged their bonus.

Every forest in this domain determines a branch: remove its fixed components, all its $A_1, A_2$ components, and one $A_3$ exactly when required. Conversely a reachable residual multiset reconstructs a forest with the displayed cost. This proves completeness and exactness of the refinement. All counts are bounded by rank 58, and each actual fiber has at most eight residual flags and 59 rank states. This also proves termination.

### C.4 The two separately bounded context families

The extra $A_1/D_5$ charge under the joint $A_9, A_2$ guard is bounded in all 759 conditional rows. The checked upper bound is $2837/8 < 377$, with the $A_{17}$ increment included. This is an upper relaxation; attainment is not claimed.

For the family containing $A_1, A_2, A_4, A_{10}$, fix the same large representatives and rank-five count. At remaining rank $R$ enumerate

$$1 \le a_1 \le R - 6, \quad 1 \le a_4 \le \left\lfloor \frac{R - a_1 - 2}{4} \right\rfloor, \quad 1 \le a_2 \le \left\lfloor \frac{R - a_1 - 4a_4}{2} \right\rfloor.$$

Apply the no-$A_2$ table to the remaining rank and add all explicit costs and context terms, including $64a_2a_4$. Permitting extra $A_1$ or $A_4$ in that table only enlarges the domain and therefore gives a valid upper bound. This has 539 coarse classes and 37 985 count choices. Refine all 14 classes whose coarse value exceeds 377 into their 77 actual fixed assignments containing $A_{10}$, and repeat the bounded loops with the actual tables. Every one of the 23 376 refined choices is at most $2861/8 < 377$. This also bounds any overlap with the first family.

The complementary calculation checks 1549 residual tables, 96 528 contextual choices, 758 838 last-item upper inequalities and 31 545 equality predecessors. Its only class above 377 has residual rank zero; its eight actual fibers are precisely those in Proposition 6. The finite loop bounds and the preceding removal argument explain why these are all the exceptions, rather than eight examples found by a search.

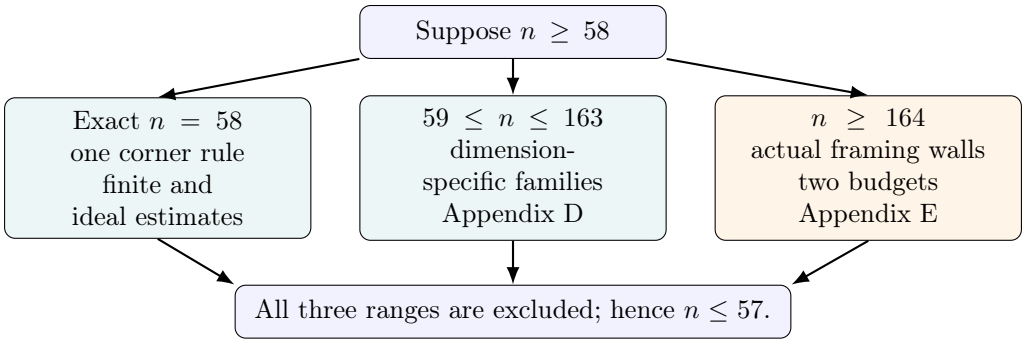



Figure 3: The higher-dimensional tail is a separate proof, not a consequence of excluding dimension58. Each branch retains its stated geometric and finite-data inputs. The arrows are logical dependencies; they do not pass to a facet assumed to inherit Coxeter angles.

# D The bounded higher-dimensional tail

The purpose of this appendix is to state the mathematical objects behind the higher-dimensional certificates. The rules change only at a small number of ranks. Their shared comparison and algorithms are more useful than the order in which those changes were discovered.

## D.1 The common comparison and terminal algorithm

Put $d = n - 1$ and

$$\beta_n = \begin{cases} n(n-6)/8 & n \text{ even}, \\ (n-1)(n-5)/8 & n \text{ odd}. \end{cases}$$

The incidence argument of Lemma 14 applies at every rank:

$$\mathcal{D}_5 \geq \beta_n v_f + \sum_{v \text{ ideal}} \kappa(v) + \Delta_n, \qquad \Delta_n > 0.$$

The weaker bound $\Delta_n \geq 5(n-1)$ suffices here. For a simultaneous rule with finite bound $U_n$ and ideal charge $C(v)$, it is enough that $U_n \leq \beta_n$ and $\kappa(v) - C(v) > 0$ at every cusp. Subtraction as in Lemma 2 then gives a contradiction, even if there are no ideal vertices.

For a cusp profile $\boldsymbol{r} = (r_1, \ldots, r_s)$ of total rank$d$, define

$$H_{\boldsymbol{r}}(t) = \prod_i (1 + \cdots + t^{r_i}) = \sum_j h_j t^j, \quad V_{\boldsymbol{r}} = \prod_i (r_i + 1), \quad G_{\boldsymbol{r}} = \sum_j h_j \min(j, d - j).$$

The normalized surplus associated with an affine-factor envelope$c$ is

$$m_c(\boldsymbol{r}) = \frac{5G_{\boldsymbol{r}} - dV_{\boldsymbol{r}}}{2V_{\boldsymbol{r}}} - \sum_i c(r_i). \tag{7}$$

The factor-fiber theorem gives $C(v) \leq V_{\boldsymbol{r}} \sum_i c(r_i)$. This multiplicity is a geometric statement; a graph pair count alone would not establish it.

Lemma 12 permits the move $(a, b) \mapsto (a - 1, b + 1)$, $1 \leq a \leq b$, whenever $c(a-1) + c(b+1) \geq c(a) + c(b)$. The normalized curvature, and hence (7), does not increase. The product of the two link vertex counts decreases, since $(a+1)(b+1) - a(b+2) = b - a + 1 > 0$. Every move raises the integer sum of squared ranks by $2(b - a + 1)$, while that sum is at most$d^2$. Thus concentration

halts. A positive terminal margin extends to every original cusp. If an absolute surplus is used, its normalized positivity is checked before multiplying by the larger original vertex count.

The terminal enumeration is the sorted-partition recursion in Section 5, with the following additional multiplicity conditions. For a linear/cap envelope $c(r) = \min(r/2, L/2)$, terminal profiles have arbitrary copies of$L$, at most one part below$L$, and at most one above$L$. For a low-only reduction, they have at most one part below$L$ and arbitrary parts at least$L$; high parts are not concentrated. In a piecewise rule the freely repeated ranks are listed below, and each other named exceptional rank occurs at most once. Adjoin each part only if these conditions survive. The remaining rank decreases at every recursive call. The unique first part of a sorted partition proves completeness and uniqueness.

For each profile compute$H$ by integer convolution and$V$ independently by its product. Check the degree, symmetry and coefficient sum; compute$G$ from its weighted sum and independently from cumulative probabilities of the sum of uniforms. Then evaluate (7) exactly. A certificate verifies equality of its profile domain with this recursion, every coefficient identity and every positive margin. An advertised minimum also carries an attaining profile. These are finite algorithms with proved domains, not a search for a favorable cusp decomposition.

Finite-vertex costs use the same typed-forest algorithm as Appendix C. The fixed counts and residual state retain every multiplicity or guard that affects a later charge. All last-item inequalities are checked, and equality predecessors are supplied whenever an exact maximum is asserted.

### D.2 The elementary families in dimensions98–163

A feature is a node of degree at least three or an endpoint of a nonsimple bond, including a parallel bond. Distances are graph distances. A condition on a whole component always concerns the whole actual endpoint diagram.

For $98 \le n \le 108$, finite weights are one at distance at most9 or in a whole component of order at most17. Their total is at most$9n$. At a cusp, pay one at distance at most8, in a whole component of order at most17, or at distance9 if a mark is within14 steps of a feature. These are alternative conditions for one unit, not separate charges. For an affine component of order$m \ge 18$, the exact counts are

| | |
|---|---|
| $A$ | $8m$ |
| $C$ | $\min(9m-45, 8m-4)$ |
| $B$ | $\min(9m-44, 8m-2)$ |
| $D$ | $\min(9m-43, 8m)$. |

For example, a $C$ spine contributes$8m-36$ short pairs; the extra distance-nine starts occupy the first16 and last16 available positions, giving $\min(m-9, 32)$. In$B$ the spine windows have lengths15 and16 and the leaves add two pairs. In$D$ both windows have length15 and the leaves add four pairs. The union of the first$a$ and last$b$ positions in a list of length$h$ has size$\min(h, a+b)$; this also covers overlapping windows. Thus these are all-rank counts. Orders at most17 cost at most $\binom{m}{2} \le 8m$, so $c(r) = \min(8, r/2)$.

For $k = 9, \dots, 13$, use unit weights at distance at most$k$ or in whole components of order at most17, at both finite and ideal vertices. The finite bound is$kn$ and $c(r) = \min(k, r/2)$. For $k = 14$ the distance condition alone suffices. The complete local replacement theorem for these distance families is the core-routing theorem of Appendix A, with the corresponding retained short-path threshold. It includes overlapping cores and arbitrary additional supporting whites. The affine graph count does not prove that local theorem.

The terminal algorithm above, with$L = 2k$, gives the following exact positive minima over the

indicated dimensions and their entire terminal domains:

| $n$ | $U_n$ | $c(r)$ | $\min m_c$ |
|---|---|---|---|
| 98:108 | $9n$ | $\min(8, r/2)$ | 112101/1419857 |
| 109:116 | $9n$ | $\min(9, r/2)$ | 45873/2476099 |
| 117:126 | $10n$ | $\min(10, r/2)$ | 145234/4084101 |
| 127:136 | $11n$ | $\min(11, r/2)$ | 40603/559682 |
| 137:146 | $12n$ | $\min(12, r/2)$ | 172096/1953125 |
| 147:156 | $13n$ | $\min(13, r/2)$ | 830407/9565938 |
| 157:163 | $14n$ | $\min(14, r/2)$ | 1476616/20511149 |

In every row $\beta_n > U_n$. This proves the exclusion of98–163.

### D.3 Labelled rules for dimensions66–97

We give the marked rules, not just their component sums. The finite supplement tables used here are lists of unordered node pairs. They have a mathematical generation algorithm stated below.

Number an affine component of order$m$ as follows. Type$A$ is the $m$-cycle. Type$C$ is the path$0, \dots, m-1$, with label4 on the first and last bonds. Type$B$ has spine$0, \dots, m-3$, its last bond labelled4, and simple leaves$m-2, m-1$ at0. Type$D$ has spine $0, \dots, m-5$ and leaves$m-4, m-3$ at0 and$m-2, m-1$ at$m-5$. All other bonds are simple and all unlisted pairs are orthogonal. Finite$A_r, B_r$ use the path$0, \dots, r-1$, with only the last $B$ bond doubled; finite$D_r$ has spine$0, \dots, r-3$ and leaves $r-2, r-1$ at$r-3$. Positions are zero-based.

Let$q_b$ pay one at distance at most$b$, or at distance $b+1, \dots, 9$ with a mark within$b$ of a feature. Let$P_b(m, X)$ be its preferred-image completion in affine type$X$, with the validated removals described below. At all other marked pairs the weight is zero. The small ideal tables are:

| $n$ | all pairs paid | exceptional tables | larger orders |
|---|---|---|---|
| 92:97 | $m \le 16$ | $P_7(17, X)$ | $q_7$ |
| 87:91 | $m \le 15$ | $P_7(16, X), P_7(17, X)$ | $q_7$ |
| 81:86 | $m \le 14$ | $P_6(15, X), P_6(16, X), P_6(17, X)$ | $q_6$ |
| 78:80 | $m \le 13$ | $P_5(14, X), \dots, P_5(17, X)$ | $q_5$. |

The totals, in the order$A, B, C, D$, are

| | | | | |
|---|---|---|---|---|
| $P_7(17)$ | 119 | 111 | 108 | 127 |
| $P_7(16)$ | 112 | 107 | 103 | 111 |
| $P_6(15)$ | 90 | 100 | 95 | 96 |
| $P_6(16)$ | 96 | 107 | 103 | 111 |
| $P_6(17)$ | 102 | 111 | 108 | 127 |
| $P_5(14)$ | 70 | 87 | 84 | 82 |
| $P_5(15)$ | 75 | 100 | 95 | 96 |
| $P_5(16)$ | 80 | 107 | 103 | 111 |
| $P_5(17)$ | 85 | 111 | 108 | 127. |

These totals check the tables; they do not define their marked entries.

Write$R_9$ for unit finite weight at distance at most9 or whole rank at most17. For81–97 add a half between any mark in a whole component of rank at most4 and a mark in$D_{12}, \dots, D_{16}$ at its fork center or short leaf, or in$A_{13}, A_{15}$ at its middle. The full cost is at most$9n$ for$n \le 108$. This slightly enlarges the earliest cross rules, preserving every local payment.

Write$R_8$ for unit weight at distance at most8, at distance9 with a whole-component leaf mark, or in whole rank at most17. Let$F_8$ add a half between any mark in whole rank at most5 and a mark of $D_{11}, \dots, D_{16}$ at its fork center or short leaf, of $A_{13}, A_{15}$ at its middle, or of$A_{17}$ at position5 or11. Also add a half between an$A_4$ endpoint and the unique long-arm neighbor of the$D_{13}$ fork center. Count each marked pair once. The complete finite bound is$8n$ for$n \le 80$.

For78–80 use$F_8$ enlarged from$R_8$ to$R_9$. The added nonleaf distance-nine pairs number$r - 11$ in a rank$r \ge 18$ path and$r - 12$ in$D_r$, at most$r$. Ranks at most17 already pay all pairs. Thus its cost is at most$9n$. The$D_{11}$ and$A_{17}$ coupons must be retained: the ideal-five cycle replacements can require them. For76–77 use$F_8$ itself. For70–75 also upgrade the half for $A_1/D_{12}$ at a short leaf to1, and add a half for either$A_2$ mark paired with the short terminal of$B_{10}$ incident to its doubled bond. The bound remains$8n$ for$n \le 75$.

For69 additionally delete the finite$A_{17}$ endpoint pair, upgrade$A_1/D_{11}$ short-leaf pairs to1, and pay1 for$A_1/A_{11}$ or1/2 for$A_2/A_{11}$ at the$A_{11}$ middle. Its finite maximum at69 is544. For67–68 replace the rank17 same-component tables by$T_{17}(A), T_{17}(B), T_{17}(D)$ of costs114,105,127 defined below; the maxima become504,512. For66 also change short-leaf distance-nine pairs in whole$D_r$, $r \ge 20$, from1 to1/2, add a half between whole rank at most5 and a short leaf of$D_{10}$, and upgrade$A_2/D_{11}$ short-leaf halves to1. Its maximum is495. All these are actual whole-component conditions. A representative count may allow$B_{17}$ the larger$D_{17}$ cost127, but may not identify the coupon-bearing$A_{17}$ with those types.

For the ideal weights below78 start with$P_5$ at orders14–17, all pairs at orders at most13, and$q_5$ at orders at least18. Apply the following setting operations for the listed dimensions:

| $n$ | Patch |
|---|---|
| 66–77 | $D$,order17: set$(1, 11)$ to0. |
| 66–75 | $B$,order15: set$(0, 11), (11, 13), (11, 14)$ to0. |
| 66–74 | Use the order13 tables below; set$H_{17}$ to1/2. |
| 66–71 | $D$,order16: set$H_{16}$ to1/2. |
| 66–70 | $B$,order14: set$(0, 10)$ to0; $B$,order15: set$(1, 11)$ to0. |
| 66–69 | Use the order12 tables below. |
| 66 | $B$,order12: set$(9, 10), (9, 11)$ to0; $C$,order12: set$(0, 10), (1, 11)$ to0; $B$,order14: set$(10, 12), (10, 13)$ to0. |

Here

$$\begin{aligned} H_{17} = \{&(0, 10), (2, 12), (0, 11), (1, 12), (1, 15), (1, 16),\\ &(11, 13), (11, 14), (2, 15), (2, 16), (10, 13), (10, 14)\},\\ H_{16} = \{&(0, 10), (1, 11), (1, 14), (1, 15), (10, 12), (10, 13),\\ &(2, 14), (2, 15), (9, 12), (9, 13)\}. \end{aligned}$$

At order13 use cycle distance at most5 in$A$; all pairs except $(0, 10), (10, 11), (10, 12)$ in$B$; all except $(0, 10), (0, 11), (0, 12), (1, 12), (2, 12)$ in$C$; and all except $\{9, 10\} \times \{11, 12\}$ in$D$. The costs are65,75,73,74. At order12 use cycle distance at most5 in$A$; all pairs with halves at$(9, 10), (9, 11)$ in$B$; all except$(0, 11)$ in$C$; and all except$\{8, 9\} \times \{10, 11\}$ in$D$. The costs before the66 patch are60,65,65,62 and after it 60,64,63,62. Orders at most11 remain wholly paid. Each marked set is invariant under the labelled target's automorphisms; equivalently apply it under every identification with that target.

### D.4 Finite-table generation and verification

A preferred record consists of two labelled minimal cores, their induced overlap, the two marked black pairs, the selected connected corner component$J$, its two marks, and its proved whole-completion order bound$h \le 17$. Appendix A defines the admissible core domain and its selected routes. For every record with$h \ge m$ and$|J| \le m$, enumerate all induced labelled injections into the target of order$m$. Assign an unassigned pattern node and try every unused target node having the correct labels to all nodes already assigned, including zero labels. The recursion has depth at most17 and branching at most17. Each injection has a unique branch, so induction proves termination, completeness and absence of missed orientations. Take the union of marked images, not their multiplicity.

The table$P_b(m, X)$ is this union with$q_b$, followed by the validated cycle removal above distance$b$ and, in the applicable$D$ targets, removal of

$$\{p : \operatorname{dist}(p) \ge m - 4\} \ \cup \ \{\text{the pair of degree-three nodes}\}.$$

The latter has nine pairs. The finite supplement contains the labelled pattern records and paid-set records$(b, m, X, P)$, where$P$ is a list of distinct unordered pairs$0 \le i < j < m$. Missing entries mean zero. The preferred-image algorithm and the setting operations above therefore specify the actual weights without executing an import.

The initial ideal-five domain has949 overlap records,571 with bound at least14. The distance-eight domain has2614 records,1153 with bound at least14; its entire marked image union lies in those same paid or validated-removal sets. It changes no table. Ideal-six uses437 eligible records; ideal-seven uses260 at order16 and80 at order17. These numbers are consistency checks, not completeness proofs in place of the stated domains.

For$T_{17}$ take the union of$R_8$'s short/leaf-nine pairs, all bound17 selected marked images, and all marked images whose retained replacement is a unit in the complete literal-interface ledger. Then delete the$A_{17}$ endpoint pair by its separate validated transfer. It occurs in neither the basic nor the selected high-pattern union; all literal totals are recounted after deletion. This finite definition is stronger than merely recording the costs114,105,127. Equivalently its complete literal description is the basic short/leaf-nine set together with

$$\begin{array}{l|l} A_{17} & (2,11),(2,12),(2,13),(2,14),(3,12),(3,13),\\ & (3,14),(4,13),(4,14),(5,14),(0,13),(3,16)\\ B_{17} & (2,11),(3,12),(4,13), \end{array}$$

with unit weights, while$D_{17}$ pays every pair except

$$(0,10),(0,11),(1,10),(1,11),(1,12),\\ (1,13),(1,14),(1,15),(1,16).$$

This explicit alternative fixes every marked entry, including old zeros.

The supplement verifier reconstructs all target graphs and all induced injections, compares the resulting pair sets, checks every patch entry and automorphism image, and sums every actual component and cross-role cost. It then checks the Bellman upper inequalities for the complete typed forest domain. Its correctness is the preceding union argument and induction on residual rank. Its halting follows from the explicit order17 target bound and the finite rank limit of the particular interval. The geometric proof separately shows that every removed preferred or literal image has a retained actual replacement and that every added-white contact is covered; neither the pair-list verifier nor the forest maximum proves that statement.

### D.5 Envelopes and recorded bounded calculations

For ideal-five factors of order$m \geq 18$, the all-rank counts are

$$\begin{aligned} p_A(m) &= 5m, \\ p_C(m) &= 5m - 15 + \sum_{k=6}^{9} \min(m-k, 14), \\ p_B(m) &= 5m - 6 + \sum_{k=8}^{11} \min(m-k, 13), \\ p_D(m) &= 5m + 3 + \sum_{k=10}^{13} \min(m-k, 12). \end{aligned}$$

The same initial/final-window argument used above proves these formulas including overlap. Put$g(r) = \max_X p_X(r+1)/(r+1)$. Their five linear branches have slopes9,8,7,6,5 and give $g(r) \leq 163/23$, with equality at$r = 22$, and $g(r) = (5r+56)/(r+1)$ for$r \geq 24$.

For66–80 use$r/2$ through the stated low rank, the following successive values up to rank16, and the constant$163/23$ at every rank at least17:

| $n$ | linear through | successive exceptional values |
|---|---|---|
| 66 | 10 | $16/3,\ 75/13,\ 6,\ 32/5,\ 107/16,\ 120/17$ |
| 67:69 | 10 | $65/12,\ 75/13,\ 43/7,\ 32/5,\ 107/16,\ 120/17$ |
| 70 | 11 | $75/13,\ 43/7,\ 32/5,\ 107/16,\ 120/17$ |
| 71 | 11 | $75/13,\ 87/14,\ 97/15,\ 107/16,\ 120/17$ |
| 72:74 | 11 | $75/13,\ 87/14,\ 97/15,\ 111/16,\ 120/17$ |
| 75 | 12 | $87/14,\ 97/15,\ 111/16,\ 126/17$ |
| 76:77 | 12 | $87/14,\ 20/3,\ 111/16,\ 126/17$ |
| 78:80 | 12 | $87/14,\ 20/3,\ 111/16,\ 127/17.$ |

For81–86 the envelope is linear through13, then $20/3, 111/16, 127/17$, then constant$95/13$. For87–91 it is linear through14, then$7, 127/17$, then constant$217/29$. For92–97 it is$\min(15/2, r/2)$.

The freely repeated terminal ranks are respectively

| $n$ | repeated ranks | at most once |
|---|---|---|
| 66 | $10, 11, 12, 14, 15, 16, 17$ | $13;\ \text{one} < 10,\ \text{one} > 17$ |
| 67:69 | $10, 11, 13, 14, 15, 16, 17$ | $12;\ \text{one} < 10,\ \text{one} > 17$ |
| 70:74 | $11, 13, 14, 15, 16, 17$ | $12;\ \text{one} < 11,\ \text{one} > 17$ |
| 75:80 | $12, 14, 16, 17$ | $13, 15;\ \text{one} < 12,\ \text{one} > 17$ |
| 81:86 | $13, 16, 17$ | $14, 15;\ \text{one} < 13,\ \text{one} > 17$ |
| 87:91 | $14, 16, 17$ | $15;\ \text{one} < 14,\ \text{one} > 17.$ |

The special moves$2j \mapsto (j-1, j+1)$ have cost gains $11/65$ at$j = 13$ for66; $23/1092$ at$j = 12$ for67–69; $19/182$ at$j = 12$ for70; $16/91$ at$j = 12$ for71–74; and at$j = 13, 15$, respectively, $(4/105, 7/2040)$ for75, $(5/21, 83/408)$ for76–77, and$(5/21, 107/408)$ for78–80. For81–86 the gains at14,15 are$5/48, 107/408$; for87–91 the gain at15 is$8/17$. All are positive. The rank12 duplicate move at66 is deliberately not used: its gain would be negative.

The recorded exact calculations are:

| $n$ | $U_n$ | $\min m_c$ |
|---|---|---|
| 66 | 495 | 13367/495924 |
| 67:68 | 504, 512 | 851/165888 |
| 69 | 544 | 3641/13608 |
| 70 | $8n$ | 1819/51840 |
| 71 | $8n$ | 23641/338688 |
| 72:74 | $8n$ | 6875/49152 |
| 75 | $8n$ | 63/74698 |
| 76:77 | $8n$ | 11615/448188 |
| 78:80 | $9n$ | 409611/2171546 |
| 81:86 | $9n$ | 940675/11294304 |
| 87:91 | $9n$ | 146417/911250 |
| 92:97 | $9n$ | 10121/4194304. |

For an interval the minimum is taken over all its dimensions and all terminal profiles. The finite comparisons satisfy$U_n \le \beta_n$. Consequently the single comparison at the start of the appendix excludes66–97.

### D.6 The complete ambient59–65 layer table

Write$R_N$ for the rule used at ambient dimension$N$. Starting from the literal ambient66 rule above, apply the following assignments in the order65 down to59. An assignment *sets* a weight; it is not a repeated debit. Other entries inherit. In the cross table later rows replace earlier ones, not add to them. There are no context guards until$R_{59}$. Put $E_r = \{0, r-1\}$ and$P_k(A_r) = \{k-1, r-k\}$.

Finite features in this subsection include endpoints, branch nodes and endpoints of nonsimple bonds in the whole component. Let$\delta$ be distance to this set. Outside the small-rank overrides below, the complete changes to the basic distance rule are

| $N$ | distance6 | distance7 | distance8 |
|---|---|---|---|
| 65 | unchanged | unchanged | 1/2 if both depths $\ge 17$ |
| $64, 63, 62$ | unchanged | unchanged | 0 if both depths $\ge 11$ |
| 61 | unchanged | unchanged | 0 if both depths $\ge 8$ |
| 60 | unchanged | 0 if both depths $\ge 6$ | 0 if both depths $\ge 8$ |
| 59 | 0 if both depths $\ge 6$ | 0 if both depths $\ge 6$ | 0 if both depths $\ge 8$. |

In particular the ambient59 distance-eight depth is8, not the depth-six value of the final ambient58 rule.

The finite supplement has complete tables $\mathsf{F}_{N,X,r}$, keyed by ambient layer, finite type and rank. They are lists of records$(i, j, w_{ij})$ with$0 \le i < j < r$ and exact rational weights; absent pairs mean zero. A retained-pair list instead means its unit indicator. A changed-pair list is only a patch, never a full table. Target graphs and all their zero labels must agree with the canonical labels. These are finite mathematical arrays, with exact selectors specified in the supplement receipt; one does not call a producer to evaluate a corner. The following assignments either print the full patch or name a complete $\mathsf{F}$ table.

At65 set$A_{16}$ pairs$(0,14), (0,15), (1,14), (1,15)$ to zero. Set to zero in$B_{16}$

$$(0,11), (0,12), (0,13), (0,14), (0,15), (1,12), (1,13), (1,14), (1,15),$$
$$(2,13), (2,14), (2,15), (3,14), (3,15), (4,15), (5,15).$$

Halve$D_{16}$ pairs$(1,12),(1,13),(1,14),(1,15)$ and$D_{17}$ pairs$(2,13),(2,14),(2,15),(2,16)$. Their costs are $A_{16}/B_{16}/D_{16} = 116/104/118$ and $A_{17}/B_{17}/D_{17} = 114/105/125$. At64 set$D_{16}(0,11),(0,12)$ to zero and restore $D_{17}(1,12),(1,13)$ to one. At63 set to zero $A_{16}(0,12),(3,15)$, $D_{16}(0,10),(0,13)$ and $D_{17}(1,12),(1,13)$.

At62 halve$A_{15}(0,13),(1,14)$ and$D_{15}(0,13),(0,14)$, delete$B_{15}(0,14)$, halve$A_{16}(0,11),(4,15)$ and $D_{16}(2,13),(2,14),(2,15)$, and set $A_{17}(1,12),(4,15)$ to halves. In$D_{17}$ set$(0,12),(0,13)$ to zero and $(0,14),(0,15),(0,16),(4,14),(4,15),(4,16)$ to halves.

At61 replace the finite tables for $A_{15}, B_{15}, D_{15}, A_{16}, D_{16}, D_{17}$ by their complete $\mathsf{F}_{61,X,r}$ arrays; all earlier zeros and halves remain part of those arrays. At60 replace the three rank15 arrays by $\mathsf{F}_{60,X,15}$. For$A_{16}, A_{17}$ take the complete full-payment exchange arrays and set respectively $(0,11),(4,15)$ to1/8 and$(1,12),(4,15)$ to7/8. For$D_{16}, D_{17}$ take the complete eighth-balance arrays and set respectively$(0,14),(0,15)$ to7/8 and$(1,15),(1,16)$ to1/8. These setting operations, not the arrays' earlier$5/8, 3/8$ versions, define$\mathsf{F}_{60}$.

At59 use complete$\mathsf{F}_{59}$ arrays for all three rank14 types,$A_{15}, D_{15}, D_{16}, A_{17}$. Relative to$R_{60}$, set $D_{17}(2,12)$ to5/8, $(2,14)$ to0, and$(3,15),(3,16)$ to0. The two disjoint$D_{17}$ patches are accumulated; their separate earlier whole arrays must not overwrite each other. The base costs, useful for checking these full marked arrays, are

| $N$ | $A_{15}$ | $B_{15}$ | $D_{15}$ | $A_{16}$ | $D_{16}$ | $D_{17}$ |
|---|---|---|---|---|---|---|
| 62 | 104 | 104 | 104 | 113 | 225/2 | 120 |
| 61 | 103 | 103 | 103 | 111 | 111 | 119 |
| 60 | 101 | 97 | 201/2 | 441/4 | 443/4 | 467/4 |
| 59 | 100 | 97 | 100 | 441/4 | 439/4 | 919/8. |

At59 the rank14 costs are$89, 87, 181/2$ in$A, B, D$, and the$A_{17}$ cost is$451/4$. These sums do not define the marked entries.

Here is the entire accumulated ordinary cross-role update table. Every component named is whole; a small rank-five allowance includes all irreducible types where the broad rule allows them. An entry with a product of mark sets applies to every pair in that product. All unlisted entries remain their earlier values.

| $N$ | First marked component | Second marked component | Weight |
|---|---|---|---|
| 65 | $A_1$ or$A_2$, all | $D_{14}$,node10 | 1/2 |
| 64 | $E_6$,long ends | $D_{11}$,short leaves | 1/2 |
| 64 | $A_1$ | $B_{10}$,terminal9 | 1/2 |
| 63 | $A_4, E_4$ | $A_{12}, E_{12}$ | 1 |
| 63 | $A_4, E_4$ | $A_{13}, P_2$ | 1 |
| 63 | $D_5$,long0 | $A_{12}, P_2$ | 1 |
| 63 | $A_2$,all | $A_{13}, E_{13}$ | 1/2 |
| 63 | $A_2$,all | $A_{14}, P_2$ | 1/2 |
| 63 | $E_6$,long ends | $A_{11}, P_2$ | 1 |
| 63 | $A_2$,all | $B_9$,terminal8 | 1/2 |
| 63 | $A_2$,all, or$A_3$,middle1 | $D_{10}$,center7 | 1/2 |
| 62 | $A_2$,all, or$A_3$,middle1 | $D_{10}$,center7 | 1 |
| 62 | $E_6$,long ends | $D_{11}$,short leaves | 1 |
| 61 | $A_2$,all | $A_{13}, E_{13}$ | 1 |
| 61 | $A_1$ | $A_{13}, E_{13}$ | 1/2 |
| 61 | $A_2$,all | $A_{14}, P_3$ | 1/2 |
| 61 | $A_4, E_4$ | $A_{12}, P_3$ | 1/2 |
| 61 | $A_1$ | $B_{10}$,terminal9 | 1 |

| $N$ | First marked component | Second marked component | Weight |
|---|---|---|---|
| 60 | $A_2$,all | $D_{13}$,short leaves | 1 |
| 60 | $A_4, E_4$, or$A_5, P_2$ | $D_{11}$,short leaves | 1 |
| 60 | $D_5$,short leaves3,4 | $D_{12}$,center9 or leaves10,11 | 1 |
| 60 | $D_5$,long0 | $D_{12}$,short leaves10,11 | 7/8 |
| 60 | $D_5$,long0 | $A_{11}, P_2$ | 1 |
| 60 | $A_4, E_4$ | $B_9$,terminal8 | 1/2 |
| 60 | $D_5$,long0 | $D_{11}$,short leaves9,10 | 1 |
| 60 | $A_1$ | $B_{11}$,node9 | 1/2 |
| 60 | all marks, any rank2–5 component | $D_{16}$,center13 or leaves14,15 | 0 |
| 59 | $A_2$,all | $A_{13}$,middle6 | 7/16 |
| 59 | $A_2$,all | $D_{13}$,center10 | 7/16 |
| 59 | $A_1$ | $A_{13}$,middle6 | 9/16 |
| 59 | $A_1$ | $D_{13}$,node9 | 1/2 |
| 59 | $D_5$,long0 | $D_{12}$,short leaves10,11 | 1/2 |
| 59 | $D_5$,long0 | $A_{12}, E_{12}$ | 3/8 |
| 59 | $A_1$ | $D_5$,long0 | 3/8 |
| 59 | $D_5$,short leaves3,4 | $A_{12}, P_3$ | 1 |
| 59 | $A_1$ | $A_{14}, P_3 \cup P_4$ | 1/2 |
| 59 | $A_1$ | $A_{15}, P_3$ or$A_{16}, P_4$ | 1/2 |
| 59 | $A_1$ | $D_{14}$,node9 or$D_{15}$,node11 | 1/16 |
| 59 | $A_1$ | $A_{12}$,nodes5,6 | 1 |
| 59 | $A_1$ | $B_{12}$,node9 | 1/2 |
| 59 | $A_1$ | $A_{12}, E_{12}$ or$A_{13}, P_2$ | 1/4 |
| 59 | $A_4$,internal1,2 | $A_{13}, P_2$ | 1/2 |
| 59 | $A_4, E_4$ | $A_{11}, E_{11}$ | 1/2 |

The$D_5$ long-end/$D_{12}$ center remains its broad half: the7/8 at60, and its replacement half at59, concern only the short$D_{12}$ leaves. The full$D_5/D_{12}$ costs are therefore 45/4 and21/2. The$D_5/D_{11}$ cost is17/2 and the$D_5/A_{11}$ cost is2. These distinctions matter to which actual corner can pay. The59$A_1/D_5$ entry is unconditional; the later58 guards are not imported. There is no$A_1/A_{16}$ endpoint quarter here, and the $A_4/A_{12}$ third-node value at59 is still a half.

At59 add exactly two Boolean context roles: a marked pair in two distinct$A_2$ components receives1/8 if an unmarked$A_{10}$ and two unmarked$A_1$ components are present; an$A_2$ mark paired with the$D_5$ long end receives1/2 if unmarked$A_1, A_9$ components are present. Multiple witnesses do not multiply a payment. Their forest totals are

$$\tfrac{1}{4}a_2(a_2-1)[a_1 \geq 2,\ a_{10} > 0] + a_2 d_5[a_1 > 0,\ a_9 > 0].$$

There are no other context terms, in particular no rank15-presence same-component guards.

For ideal vertices introduce complete supplement arrays $\mathsf{I}_{N,m,X}$ with the same pair schema and zero default. At65 use the order12–17 arrays of the short-seven small-factor interface. This is a replacement, not the assumption that the older66 order15 tables already agree. Orders at most11 are all unit, orders at least18 use$q_5$. At64 halve order11 cycle distance-five pairs, $B$ pairs$(8,9),(8,10)$, the$C$ endpoint pair$(0,10)$, and$D$ pairs$\{7,8\} \times \{9,10\}$; also halve$D$,order17,$(3,15),(3,16),(9,13),(9,14)$. At63 set$C$,order11,$(0,10)$ to zero; $B$,order12,$(0,9)$ to zero; and$D$,order13,$(0,8)$ to a half. At62 set that last pair to zero, set$B$,order16,$(1,11),(2,12)$ to zero, delete the four just-halved order17$D$ pairs, and set $B$,order14,$(9,12),(9,13)$ and$C$,order14,$(1,12)$ to zero. At61 set$B$,order11,$(8,9),(8,10)$ and$C$,order11, $(0,9),(1,10)$ to zero, and set$D$,order16,$(0,9),(2,11)$ to halves.

At60 use complete$\mathsf{I}_{60}$ tables for the type/order pairs

$$(B,12),(C,12),(B,13),(C,13),(D,13),(C,14),$$
$$(B,15),(D,15),(B,16),(D,16),(D,17).$$

The order17$D$ table is the final rebalance table: its two center pairs$(0,10),(2,12)$ are restored to halves, and $(2,15),(2,16),(10,13),(10,14)$ have weight1/8. At59 use complete$\mathsf{I}_{59}$ replacements for

$$(B,13),(C,13),(B,14),(C,14),(D,14),$$
$$(B,15),(D,15),(C,16),(D,16),(D,17).$$

Equivalently the latter ideal rule is precisely the intrinsic58 table of Appendix B, except that every order10 pair has weight1. This is checked against the complete snapshots, not inferred from family totals. Orders at most10 remain wholly unit throughout59–65 and orders at least18 keep$q_5$.

The full finite and ideal arrays above are retained mathematical supplement tables, not descriptions by component cost. Their receipt keys specify type, order/rank and canonical labelled graph, with all edges and nonedges checked. Each finite snapshot is a complete pair-domain replacement, or is followed by the explicitly printed patch. Each array is bound by the corresponding immutable composition's source manifest. The supplement verifier selects by these keys, reads exact rational records, checks the pair domain and default, and applies the stated setting operations. All loops have at most$\binom{17}{2}$ entries per small table, and there are seven layers; hence evaluation halts and determines every marked weight. It is distinct from the proof that an actual face reaches those marked roles.

For reference the small-affine family maxima at orders10–17 are

| $N$ | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|
| 65 | 45 | 55 | 64 | 74 | 84 | 94 | 107 | 119 |
| 64 | 45 | 109/2 | 64 | 74 | 84 | 94 | 107 | 118 |
| 63 | 45 | 54 | 63 | 147/2 | 84 | 94 | 107 | 118 |
| 62 | 45 | 54 | 63 | 73 | 83 | 94 | 106 | 116 |
| 61 | 45 | 53 | 63 | 73 | 83 | 94 | 105 | 116 |
| 60 | 45 | 53 | 62 | 145/2 | 82 | 92 | 103 | 229/2 |
| 59 | 45 | 53 | 62 | 143/2 | 80 | 367/4 | 411/4 | 227/2. |

Divide an entry by its order to obtain$c(r)$ at rank$r = m - 1$. For59–62 use the exact$g(r)$ above rank16, with low-nine-only concentration. For63–65 use the constant163/23 above rank16; the freely repeated ranks are9–17 at63,64 and10–17 at65, with one lower and one higher part. The earlier all-rank formula gives the remaining entries. The exact arithmetic interface is

| $n$ | $U_n$ | $\min m_c$ | $\min V m_c$ |
|---|---|---|---|
| 59 | 783/2 | 1/340 | 91/2 |
| 60 | 404 | 31/6000 | 54 |
| 61 | 420 | 3/250 | 64 |
| 62 | 434 | 2741/371712 | 73 |
| 63 | 449 | 3505/201344 | 139/46 |
| 64 | 464 | 2385/146432 | 240/23 |
| 65 | 479 | 61/1936 | 445/23. |

All$U_n \le \beta_n$, so the common comparison excludes59–65. The local proof uses the same core domain, arbitrary-support seals and actual-corner priorities as Appendix A, but with each layer's own marked

table. Every newly reduced affine corner, including a nonzero fractional reduction, is handled by its whole-face theorem first. If the actual face has no such corner, its ideal weights agree pointwise with the preceding layer. Finite sparse changes then use their distinct retained whole-component alternatives. Deep corridors use the independent shorter-path route and preserve its feature and leaf outputs. Guards are justified by physical whole-component seals before entering the finite-state count. Thus no final58 restoration or extra coupon is spent in an earlier tail rule.

# E The unbounded tail: actual framing walls and two budgets

The range$n \geq 164$ requires a proof at all ranks, not a finite list of partitions. We use Nikulin's narrow-facet theorem [5, pp. 654–655]; its unrestricted statement is also given in [6, Section 4, p. 10]. A finite-volume chamber has a facet$F$ whose actual framing hyperplanes have pairwise distance strictly less than$\operatorname{arccosh} 7$. With outward normals of square2 this implies

$$-(x, y) < 14 \tag{8}$$

for any two framing walls. This is the external narrowness input. We prove the incidence, endpoint and scale statements needed to apply it; they are not additional hypotheses on the cusp.

## E.1 Outgoing edges really give framing walls

Suppose first that$F$ contains a cusp$v$. Its horospherical link is a product of affine Coxeter simplices. Write their ranks$r_a$, with$\sum_a r_a = N = n - 1$, and normalize every old root to square2. In each factor choose two omitted nodes in advance. In the factor containing the normal$\alpha_s$ of$F$, both choices avoid$s$. The two choices need not be distinct. If that factor has affine rank one, use its other node twice; its modified target$2(r_a - 1)/9$ is zero. Deleting one node per factor gives a positive-definite support$S$ of$N$ roots and a vertex of the horospherical link. It therefore defines an actual outgoing edge in$F$. Finite volume gives a finite or ideal second endpoint.

At a finite endpoint choose any new wall$x$ terminating that edge. It and$\alpha_s$ belong to a spherical diagram, so their rank-two Gram is positive definite. At an ideal endpoint let$T_b$ be its affine factors. The positive-definite support$S$ contains at most rank$\,T_b$ nodes from each. The total rank$N$ forces equality in every factor: exactly one node of each$T_b$ is missing. If the factor containing$\alpha_s$ has rank at least2, choose its missing node as$x$; every pair in such a factor has finite angle. If it is affine$A_1$, choose the missing node of another factor, which is orthogonal to$\alpha_s$. Another factor exists in the range at issue. Thus again the pair is positive definite.

To see that this is an actual ridge, start with a strict interior timelike point$z$. Project it off the span of$x, \alpha_s$. The projection correction is a nonnegative combination of those normals, because their Gram is a positive-definite matrix with nonpositive off-diagonal entries and nonnegative inverse. Every other wall inequality stays strict, and the projected square is even more negative. It gives a point of precisely their common face. Hence$x$ is in the framing set controlled by (8). Apply the same construction to the second omission tuple and call its terminating wall$y$.

Neither$x$ nor$y$ is an old deleted cusp wall: that wall is strict along the outgoing edge away from$v$. Their old contact rows $p_j = -(x, \alpha_j)$ and$q_j = -(y, \alpha_j)$ are nonnegative, and every nonzero entry is at least1. For each affine factor let $c_a = \sum_j \delta_{aj}\alpha_j$, with$\delta_{aj} > 0$ normalized by$\min_j \delta_{aj} = 1$. These are nonzero future null vectors, since their pairing with an interior point is negative. They are all positive multiples$k_a c$ of one cusp vector$c$. Therefore

$$\sigma_a = \delta_a \cdot p = -(x, c_a), \quad \tau_a = \delta_a \cdot q = -(y, c_a), \quad q_0 = \sigma_a/\tau_a$$

have a common positive ratio. Indeed, if a row height vanished, all its nonnegative contacts with that factor would vanish; proportionality of the$c_a$ would force every factor height and every old contact to vanish. A square-two normal would then lie in the null orthogonal complement of the full cusp span, impossible.

For each endpoint the principal Gram$S+x$ or$S+y$ is positive semidefinite. Its Schur complement gives the separate budgets

$$\sum_a C_{0a} \le 2, \qquad \sum_a C_{1a} \le 2, \qquad C_{\ell a} = p_{\ell,S_a}^t G(S_a)^{-1} p_{\ell,S_a}. \tag{9}$$

The inverse is entrywise nonnegative. No bound on a deleted-node contact or on its Coxeter label has been used.

### E.2 The common-height Lorentz identity

Choose a null partner$d_0$ with$(c, d_0) = -1$, and write $x = ac + bd_0 + u$, $y = a'c + b'd_0 + u'$ in the orthogonal Euclidean quotient. Then$b = -(x, c) > 0$, $b' = -(y, c) > 0$. Since $(x/b - y/b') \perp c$ and$x^2 = y^2 = 2$,

$$\left\|u/b - u'/b'\right\|^2 = 2/b^2 + 2/(b')^2 - 2(x, y)/(bb').$$

The quotient is the orthogonal sum of the affine factor spaces. In factor$a$ put $E_a = \|u_a/\sigma_a - u'_a/\tau_a\|^2$. Since$\sigma_a = k_a b$ and$\tau_a = k_a b'$, the$k_a^2$ cancels and the preceding identity becomes

$$-(x, y) = \frac{1}{2} \sum_a \sigma_a \tau_a E_a - q_0 - q_0^{-1}. \tag{10}$$

This proves the scale cancellation for a product cusp; no commensurability or chosen horosphere is required. Let$M = \max(q_0, q_0^{-1})$ and $W_a = \min(\sigma_a, \tau_a)^2 E_a$. Then $\sigma_a \tau_a E_a = M W_a$ and$q_0 + q_0^{-1} \le 2M$.

The prescribed omission choices have targets

$$T_a = 2r_a/9 \quad \text{ordinarily}, \qquad T_a = 2(r_a - 1)/9 \quad \text{in the sole avoid-node factor.}$$

Thus$\sum T_a = (2N - 2)/9$. We next prove

$$\sum_a W_a \ge \begin{cases} (2N-2)/9 - 4 & 1 \le M \le 21/20, \\ (2N-2)/9 - 16/3 & M \ge 21/20. \end{cases} \tag{11}$$

It follows from(10) that $-(x, y) \ge M(N - 37)/9$ in the first range and $-(x, y) \ge M(N - 43)/9$ in the second. At$N = 163 + t$ these are at least$14 + t/9$ and$14 + 7t/60$, respectively. They contradict (8) for every$t \ge 0$.

### E.3 Point potentials and the complete local domain

For$r$ Euclidean coordinates put$s_k = e_1 + \cdots + e_k$. The classical point potentials and normalized kernel squares are

| | $v_k$ | $(\delta_k^2)$ |
|---|---|---|
| $B$ | $v_0 = 0,\ v_1 = e_1,\ v_k = s_k/2\ (k \ge 2)$ | $(1, 1, 4, \ldots, 4, 2)$ |
| $C$ | $v_k = s_k,$ metric one half the usual metric | $(1, 2, \ldots, 2, 1)$ |
| $D$ | $v_0 = 0,\ v_1 = e_1,\ v_k = s_k/2\ (2 \le k \le r-2)$ | |
| | $v_{r-1} = (s_{r-1} - e_r)/2,\ v_r = s_r/2$ | $(1, 1, 4, \ldots, 4, 1, 1).$ |

Write$R(i,j) = \|v_i - v_j\|^2$. For an$A$ cycle of order$m$ use $\delta = 1$ and$R(i,j) = h(m-h)/m$, $h = (i-j) \bmod m$. Ordinary omissions are$(0, \lfloor m/2 \rfloor)$ in$A$ and $(0,r)$ in$B, C, D$. Modified$B, C$ use$(r-1, 0)$ and$(1,r)$. Rotations and fork automorphisms avoid a prescribed node in $A, D$ without weakening the ordinary geometric estimate.

Exceptional diagrams use the central node0 and consecutive labels along arms$(2,2,2)$, $(1,3,3)$, $(1,2,5)$ for affine$E_6, E_7, E_8$. Their rational symmetric matrices have diagonal2 and edge$-1$. For$F_4$ use diagonal$(2,2,2,4,4)$, successive edges $(-1,-1,-2,-2)$ and scales$(1,1,1,2,2)$; for$G_2$ use

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -3 \\ 0 & -3 & 6 \end{pmatrix}, \qquad (s_i) = (1,1,3).$$

The physical Gram is$A_{ij}/\sqrt{s_i s_j}$. If$k > 0$ spans $\ker A$, put$\delta_i^2 = s_i k_i^2 / \min_j(s_j k_j^2)$. Ground node0 and solve$Au = f$, $u_0 = 0$, for $f = e_i/k_i - e_j/k_j$; then $R(i,j) = \min_l(s_l k_l^2) f^t u$. This defines every resistance by finite rational linear algebra.

Ordinary exceptional omissions are $E_6 : (2,4)$, $E_7 : (3,6)$, $E_8 : (1,6)$, $F_4 : (0,4)$, $G_2 : (0,2)$. To avoid$f$, take the first two of$(2,4,6) \setminus \{f\}$ in$E_6$; $(2,5)$ if$f \in \{4,7\}$ and$(4,7)$ otherwise in$E_7$; $(1,6)$ if$f \in \{3,7\}$ and$(3,7)$ otherwise in$E_8$; $(0,4)$ if$f \in \{1,3\}$ and$(1,3)$ otherwise in$F_4$; and the other two nodes in$G_2$. Modified$B_2$ has matrix $\left(\begin{smallmatrix} 2 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 4 \end{smallmatrix}\right)$, scales$(1,1,2)$ and omissions$(0,1)$, avoiding node2.

The next enumeration is for the *residual* product after a failing factor has been selected below. Its ordered budgets are at most$3/2$ on each side. The original product still has budgets2; the failing-factor classification below is established separately and does not apply this residual cap to the selected factor. For the near range let$B = 21/20$. The finite model list consists of$A$ ranks1–16, $B, C$ ranks2–7, $D$ ranks4–17, all the ordinary and avoid-node exceptional cases, modified$B$ ranks3–7, modified$C$ ranks2–7, and modified$B_2$. For an omission$o$ a retained contact$k$ is eligible only if$\delta_k^2 R(o,k) \le 3/2$. Enumerate every subset$S$ of these nodes and retain it only if

$$c_o(S) = \sum_{k \in S} \delta_k^2 R(o,k) + \sum_{k<l \in S} \delta_k \delta_l [R(o,k) + R(o,l) - R(k,l)] \le 3/2. \tag{12}$$

The retained inverse is nonnegative, so this is a lower bound for every row with that support and magnitudes at least1. Include or omit the omitted-node contact independently on each side; included means magnitude at least1. Each full row is nonempty. The two-empty-retained case has the direct bound$W \ge T$.

For full supports$U, V$, their normalized potentials lie in two finite convex hulls. Form their positive-semidefinite signed point Gram$K$ and the two row-sum constraints$Az = (1,1)$, $z \ge 0$. An exact lower-distance certificate consists of

$$Az = (1,1), \quad Kz + A^t\lambda \ge 0, \quad z_i(Kz + A^t\lambda)_i = 0, \quad e = z^t K z = -\lambda_1 - \lambda_2.$$

Expanding the quadratic at$z$ proves that$e$ is the global minimum on the two hulls. Enumerate active subsets and solve their rational KKT equations; verify a witness by these identities. Exhaustion always halts; failure to find a witness is a failed certificate, never an omitted support.

Singular active systems may be skipped for the following reason. Choose a minimum on the product of the two simplices having the fewest positive coordinates, and restrict to that support $I$. Both side-sum constraints are nonempty and independent. If its bordered KKT matrix were singular, a nonzero null pair $(h, \nu)$ would satisfy

$$A_I h = 0, \qquad K_I h + A_I^t \nu = 0.$$

Multiplying by $h$ gives $h^t K_I h = 0$. Positive semidefiniteness implies $K_I h = 0$, and full row rank then gives $\nu = 0$. Thus $h \neq 0$ and preserves both sums. Moving along $h$ until a positive coordinate first reaches zero preserves feasibility and the minimum, contradicting the choice of support. Some minimizing active system is therefore nonsingular. Its rational solution is found by the exhaustive search. This argument proves search completeness; the displayed slack identities independently prove soundness of each supplied certificate.

Writings$s = \sum_U \delta_k$, $t = \sum_V \delta_k$, the positive row magnitudes and common ratio bound give

$$W \geq \max\{\min(s,t)^2, \max(s,t)^2/B^2\}\, e. \tag{13}$$

This covers arbitrary unbounded contact magnitudes by convex normalization, not a finite angle list.

The saved exact calculation has91 finite models,1116 retained support pairs,3630 nonempty continuous domains and1395 positive losses. Radicals in support rejection and height bounds are rounded outward to rational bounds with squared inequalities; every surviving support cost is also checked exactly. The counts are checksums. Coverage comes from the complete model list, all subsets and both omitted-node states.

The remaining ranks have no positive loss. For$B$ ordinary$r \geq 8$ the eligible retained sets are$\{1\}, \{r-1\}$; for modified$B$ they are$\{r-2, r\}, \{1\}$. The energy lower bounds are$(r-2)/4$ and $(r-3)/4$, with height-square at least$2/B^2$. For ordinary$C$ they are$\{1\}, \{r-1\}$, and for modified$C$, $\{0,2\}, \{r-1\}$; energy is at least$(r-2)/2$ and$(r-3)/2$, with height-square at least1. For$D$, $r \geq 18$, the sets are $\{1\}, \{r-1\}$ and energy is at least$(r-2)/4$ with height-square1. Omitted-node contacts may also occur. Each difference of full convex-hull points has the indicated fixed middle coordinates, of magnitude$1/2$ in$B, D$ or1 in$C$. Their number proves the energy bounds. Subtracting$T$ gives a nonnegative constant at the initial rank and a positive coefficient of the excess rank.

For an$A$ cycle, $m \geq 18$ even or$m \geq 19$ odd, cost at most$3/2$ permits only one adjacent retained node per side. Singleton full supports have separation at least$\lfloor m/2 \rfloor - 2$; the margin numerator is $m^2 + 8m - 9(4+\varepsilon)^2$ over$36m$. If a full row has two nodes, its height is at least$2/B$ and the margin numerator is $(9 - 2B^2)m^2 + (2B^2 - 72)m - 81\varepsilon$ over$9B^2 m$. Here$\varepsilon = m \bmod 2$. Substituting$m = 18 + 2u$ or$19 + 2u$, $u \geq 0$, gives nonnegative coefficients and a positive constant. This proves the tails at every rank.

### E.4 The classical unit-charge estimate at the original budgets

The residual calculation above starts only after a failing factor has been selected. To justify that selection, we first need a statement at the original endpoint budgets. In this subsection each retained row has cost at most two; neither row is restricted to the residual budget 3/2.

**Lemma 18** (Full-budget classical unit charge)**.** *For the prescribed ordinary or avoid-node omissions of a classical affine factor, let $C_0, C_1 \leq 2$ be its actual retained costs, and let $W = \min(\sigma, \tau)^2 E$. If the common height ratio is at most $B = 21/20$, then*

$$W + C_0 + C_1 \geq T,$$

*where $T = 2r/9$ ordinarily and $T = 2(r-1)/9$ in an avoid-node factor. This includes arbitrary real contact magnitudes with every nonzero contact at least one, and both possibilities for each omitted-node contact.*

*Proof.* The point potentials and kernel squares are those already displayed above. For omissions $i, j$, put

$$R = R(i,j), \qquad c_{ik} = \delta_k^2 R(i,k), \qquad \ell_{ik} = R(i,j) + R(i,k) - R(k,j).$$

The number $c_{ik}$ is the unit-contact inverse diagonal, not the actual cost of an arbitrary contact. Normalizing a full contact row by its affine height gives nonnegative convex weights $w_k$, of sum one. The omitted node itself is allowed in this convex combination; its displacement and its linear loss are zero. Write the two normalized point means as $v_i + a, v_j + b$. Direct polarization gives

$$E = R - \sum_k w_k \ell_{ik} - \sum_l z_l \ell_{jl} + \|a - b\|^2. \tag{14}$$

Every row height is at least one, so $W \geq E$. The positive retained inverse and the contact gap give

$$C_0 \geq \sum_{k \text{ retained, contacted}} c_{ik}, \qquad C_1 \geq \sum_{l \text{ retained, contacted}} c_{jl}.$$

Here all same-row cross terms are nonnegative. In particular, since $0 \leq w_k, z_l \leq 1$, a bound on each unit diagonal controls the corresponding weighted linear losses without any upper bound on a contact magnitude.

For $B, C, D$ the following table is the complete contact partition. An asterisk denotes the prescribed avoid-node omissions. The column "trivial" marks a contact for which $c_{ik} > T$; if such a contact is present, $C_0 + C_1 > T$ already proves the assertion. In every other row $\ell_{ik} \leq c_{ik}$.

| Type | $(i,j)$ | retained $k$ | $c_{ik}$ | $\ell_{ik}$ | trivial |
|---|---|---|---|---|---|
| $B$ | $(0,r)$ | $1$ | $1$ | $1$ | |
| $B$ | $(0,r)$ | $2 \leq k \leq r-1$ | $k$ | $k/2$ | |
| $B$ | $(0,r)$ | $r$ | $r/2$ | $r/2$ | |
| $B$ | $(r,0)$ | $0$ | $r/4$ | $r/2$ | yes |
| $B$ | $(r,0)$ | $1$ | $r/4$ | $r/2-1$ | yes |
| $B$ | $(r,0)$ | $2 \leq k \leq r-1$ | $r-k$ | $(r-k)/2$ | |
| $B^*$ | $(r-1,0)$ | $0$ | $(r-1)/4$ | $(r-1)/2$ | yes |
| $B^*$ | $(r-1,0)$ | $1$ | $(r-1)/4$ | $(r-3)/2$ | yes |
| $B^*$ | $(r-1,0)$ | $r$ | $1/2$ | $0$ | |
| $B^*$ | $(r-1,0)$ | $2 \leq k \leq r-2$ | $r-1-k$ | $(r-1-k)/2$ | |
| $B^*$ | $(0,r-1)$ | $1$ | $1$ | $1$ | |
| $B^*$ | $(0,r-1)$ | $r$ | $r/2$ | $(r-1)/2$ | |
| $B^*$ | $(0,r-1)$ | $2 \leq k \leq r-1$ | $k$ | $k/2$ | |
| $C$ | $(0,r)$ | $1 \leq k \leq r-1$ | $k$ | $k$ | |
| $C$ | $(0,r)$ | $r$ | $r/2$ | $r$ | yes |
| $C^*$ | $(1,r)$ | $0$ | $1/2$ | $0$ | |
| $C^*$ | $(1,r)$ | $2 \leq k \leq r-1$ | $k-1$ | $k-1$ | |
| $C^*$ | $(1,r)$ | $r$ | $(r-1)/2$ | $r-1$ | yes |
| $C^*$ | $(r,1)$ | $0$ | $r/2$ | $r-1$ | yes |
| $C^*$ | $(r,1)$ | $1 \leq k \leq r-1$ | $r-k$ | $r-k$ | |
| $D$ | $(0,r)$ | $1$ | $1$ | $1$ | |
| $D$ | $(0,r)$ | $2 \leq k \leq r-2$ | $k$ | $k/2$ | |
| $D$ | $(0,r)$ | $r-1$ | $r/4$ | $r/2-1$ | yes |
| $D$ | $(0,r)$ | $r$ | $r/4$ | $r/2$ | yes |

The ordinary ranks start at $2, 2, 4$ for $B, C, D$; the modified $B$ rows start at rank three. Empty integer ranges in the table contribute no contact. The other ordinary $C$ row follows by reversal, and the other $D$ row by the automorphisms interchanging the two ends and the fork leaves. These same fork automorphisms avoid a prescribed node while preserving the stronger ordinary $D$ estimate.

The omission distances are $r/4$ in ordinary $B, D$, $(r-1)/4$ in modified $B$, $r/2$ in ordinary $C$, and $(r-1)/2$ in modified $C$. Each is at least its target. If no trivial contact is present, applying

the table to both sums in (14) gives $W + C_0 + C_1 \geq R \geq T$. The exceptional modified $B_2$ matrix displayed earlier has $R(0,1) = 1$ and every retained inverse diagonal at least $1/2 > T = 2/9$. Thus a nonempty retained support is trivial; if both supports are empty, the omitted-only rows give $W \geq 1$. This proves all $B, C, D$ ranks. Notice that this is a coefficient-one estimate; the later coefficient-4/3 estimate alone would not prove the present lemma.

For $A$ write $m = r + 1$, $h = \lfloor m/2 \rfloor$ and $\varepsilon = m \bmod 2$. The omissions are $0, h$, and $R(i,j) = d(m-d)/m$, where $d = (i-j) \bmod m$. We first state the finite calculation at small orders with its full domain.

For each $2 \leq m \leq 17$ and each omission $o \in \{0, h\}$, enumerate all subsets of

$$\{k \neq o : R(o,k) \leq 2\}.$$

Keep $S$ exactly when its complete unit-contact cost

$$c_o(S) = \sum_{k \in S} R(o,k) + \sum_{k<l \in S} [R(o,k) + R(o,l) - R(k,l)]$$

is at most two. Pair the two retained sets, keeping pairs with $c_0(S_0) + c_h(S_1) < T$; other pairs are already trivial. Include or omit each omitted node independently to form full supports $U, V$, and require both full supports nonempty. These are exactly the possible support domains relevant to a failure: actual costs bound their unit costs below, and positive affine height excludes an empty full row. No residual cap, finite angle cutoff or bound on an omitted contact has entered this construction.

Use the Euclidean potentials

$$v_k = e_1 + \cdots + e_k - \frac{k}{m}(e_1 + \cdots + e_m).$$

Their Gram is $K_{ij} = \min(i,j) - ij/m$, and their squared distances are $R(i,j)$. For a pair $U, V$, form the signed point Gram $K$, with positive signs on $U$ and negative signs on $V$. It is positive semidefinite. A rational certificate for its closest convex means consists of $z \geq 0$, the two row-sum conditions $Az = (1,1)$, and multipliers $\lambda$ satisfying

$$Kz + A^t\lambda \geq 0, \qquad z_i(Kz + A^t\lambda)_i = 0.$$

Then $e = z^tKz = -\lambda_1 - \lambda_2$ is a global lower bound for the squared distance: for any feasible $z'$, expansion gives

$$(z')^tKz' - e = (z'-z)^tK(z'-z) + 2(z')^t(Kz + A^t\lambda) \geq 0.$$

This also covers singular Gram matrices. With $s = |U|$, $t = |V|$, the common ratio bound and the contact gap imply

$$\min(\sigma,\tau)^2 \geq H_B(s,t) := \max\{\min(s,t)^2, \max(s,t)^2/B^2\}.$$

Consequently $W + C_0 + C_1 \geq H_B(s,t)e + c_0(S_0) + c_h(S_1)$.

The recorded full-domain calculation has *580 cases*. For every one, its supplied rational witness satisfies the displayed KKT conditions and

$$H_{21/20}(|U|,|V|)e + c_0(S_0) + c_h(S_1) \geq 2(m-1)/9. \tag{15}$$

This is the finite computed proposition used here. Its verifier reconstructs all the preceding sets and both omitted-node states, matches the witness keys $(m, U, V)$, and verifies their exact rational

identities and inequalities. Within each selected $m$, the witness map rejects missing, duplicate or out-of-domain support keys. The separate coverage guard also requires one wrapper for each $m = 2, \dots, 17$ and the exact widened scalar-key set; a wrapper lookup alone would not reject redundant duplicate receipt rows. There are sixteen values of $m$, finitely many subsets of at most sixteen retained nodes, four omitted-node states per pair, and finitely many rational operations per supplied witness. Thus verification halts. Completeness is the exhaustive support construction, not the number 580. The KKT identity and the height bound prove correctness for arbitrary contact magnitudes.

Crucially, (15) is the calculation at $B = 21/20$. An earlier height window $31/30$ cannot be substituted: increasing $B$ weakens $H_B$. The same rational closest-point witnesses are used, but every height and unit-charge inequality is rechecked at the wider window.

It remains to prove the $A$ tail at all orders. For $m \geq 18$, a retained contact at cyclic distance $d \geq 3$ and $d \leq m/2$ has unit cost at least $3(m-3)/m > 2$. Hence each contacted retained node has distance one or two from its omission. Direct substitution in the resistance formula gives

$$\frac{\ell_{ik}}{c_{ik}} \leq \frac{m+\varepsilon}{m-d} \leq L_m := \frac{m+\varepsilon}{m-2}.$$

The same holds in the other row. Therefore polarization gives

$$W + C_0 + C_1 - T \geq R(0,h) - T - (L_m - 1)(C_0 + C_1) \geq R(0,h) - T - \frac{4(2+\varepsilon)}{m-2}.$$

For even $m = 18 + 2u$, $u \geq 0$, the last quantity is

$$\frac{m^2 + 6m - 304}{36(m-2)} = \frac{128 + 84u + 4u^2}{36(m-2)} > 0.$$

For odd $m = 19 + 2u$ it is

$$\frac{m^3 + 6m^2 - 457m + 18}{36m(m-2)} = \frac{360 + 1708u + 252u^2 + 8u^3}{36m(m-2)} > 0.$$

These are all-order identities, not boundary tests or extrapolation. Rotation chooses an opposite omission pair avoiding any prescribed node when $m \geq 3$, so the stronger ordinary estimate covers the modified target too. At $m = 2$ the repeated allowed omission has modified target zero, for which nonnegativity suffices. This completes the proof. □

The lemma closes the classical part of the first-failing-factor classification at the original budgets. The separate exceptional enumeration that follows is consequently sufficient: any factor with $T - W > C_0 + C_1$ must occur among its exceptional supports. Only after selecting one of those factors are the residual $3/2$ budgets and the ordered packing table applied.

### E.5 Ordered packing proves loss at most four

Each positive-loss domain is dominated by a record with no larger ordered cost floors, no smaller loss bound, and the same modified flag. The complete28-record table is:

| Type | $(C_0, C_1)$ floors | loss bound |
|---|---|---|
| ordinary $A_1$ | $(1/2, 0)$ | $2/9$ |
| ordinary $A_2$ | $(2/3, 0)$ | $4/9$ |
| ordinary $A_4$ | $(4/5, 4/5)$ | $8/9$ |
| ordinary $B_r,\ 2 \le r \le 5$ | $(0, r/4)$ | $2r/9$ |
| ordinary $D_r,\ r = 4, 5$ | $(r/4, 0)$ | $2r/9$ |
| ordinary $E_6$ | $(0, 4/3), (4/3, 0)$ | $4/3$ each |
| ordinary $E_7$ | $(1/2, 1), (1, 1/2)$ | $14/9$ each |
| ordinary $E_8$ | $(8/9, 2/3)$ | $16/9$ |
| modified $B_r,\ 3 \le r \le 7$ | $((r-1)/4, 0)$ | $2(r-1)/9$ |
| modified $E_6$ | $(0, 4/3)$ | $10/9$ |
| modified $E_7$ | $(0, 3/2), (2/3, 5/6), (5/6, 2/3)$ | $4/3$ each |
| modified $E_8$ | $(1, 1/2), (8/9, 2/3)$ | $14/9$ each |
| modified $F_4$ | $(0, 1), (1/2, 3/4)$ | $2/3$ each |
| modified $G_2$ | $(0, 1/2)$ | $2/9$ |

The verifier supplies and checks a dominating record for each of the 1395 positive domains. It does not assume the table is geometrically realizable by one chamber.

For records with indices at least$i$, ordered remaining budgets$p, q$, and at most$a$ modified records, the maximum residual loss is the maximum of0 and

$$\ell_j + F(j, p - p_j, q - q_j, a - \epsilon_j), \qquad j \ge i,$$

over admissible records. Each has$p_j + q_j \ge 1/2$, so the recursion has length at most$\lfloor 2(p+q) \rfloor$. All costs have a common denominator, hence the state set is finite. Induction on remaining length proves completeness; checking every upper inequality and an equality predecessor proves the recorded maxima

$$F(0, 10/9, 4/3, 1) = 20/9, \qquad F(0, 1, 3/2, 1) = F(0, 3/2, 1, 1) = 22/9.$$

One additional complete exceptional check is needed. If $T - W > C_0 + C_1$, then it is an ordinary$E_8$ with cost floors $(8/9, 2/3)$ and target$16/9$, or an ordinary$E_7$/modified$E_8$ with floors$(1/2, 1)$ or$(1, 1/2)$ and target$14/9$. Its finite algorithm enumerates every grounded exceptional support below the trivial threshold$C_0 + C_1 < T$, including all inverse cross terms, and checks the resulting convex-quadratic bounds. There are15 candidate exceptional supports. Two ordinary$E_7$ same-row supports at nodes4,7 are excluded by exact convex certificates, in both omitted-node states, giving$W + 3/2 > T$. For the omissions $(3, 6)$ the four ordered full-support pairs are

$$((4, 7), (6)), \quad ((4, 7, 3), (6)), \quad ((3), (4, 7)), \quad ((3), (4, 7, 6)).$$

The coverage guard requires these four distinct keys, not merely four individually valid certificates. The remaining13 have exactly the stated floors. This support check is separate from the residual domination table.

If no factor fails, the two budgets already bound total loss by4. Otherwise choose any failing factor. An$E_8$ leaves budgets at most$(10/9, 4/3)$, and its own loss at most$16/9$ plus the residual $20/9$ is4. In the other case the residual is at most$22/9$ and the own loss at most$14/9$, again4. All additional failures are included in this residual calculation. Permitting one modified residual record even if the selected factor was modified only enlarges the upper bound. This proves the first line of (11).

## E.6 The four-thirds estimate for the other height range

For the same omission choices the all-rank local inequality is

$$W + \frac{4}{3}(C_0 + C_1) \geq T.$$

To prove it, write the two normalized point potentials as $v_i + a, v_j + b$. With convex contact weights$w_k, z_k$,

$$E = R_{ij} - \sum_k w_k(R_{ij} + R_{ik} - R_{kj}) - \sum_k z_k(R_{ij} + R_{jk} - R_{ki}) + \|a - b\|^2.$$

A nonzero contact has magnitude at least1, every convex weight is at most1, and the nonnegative retained inverse makes its cost dominate the sum of unit diagonal costs. Also$W \geq E$, since both row heights are at least1.

For $B, C, D$, use the coefficient-one argument in Lemma 18. That part of its proof uses only that both row heights are at least one, and hence is independent of the near-range ratio bound. Its complete contact table also covers modified $B_2$. Increasing the cost coefficient to $4/3$ preserves the inequality. The $A$ and exceptional factors in this height range require the following separate arguments.

For $A$ put $C = C_0 + C_1$ and $T = 2(m-1)/9$. If $C \geq 3T/4 = (m-1)/6$, nonnegativity of $W$ proves the claim. Only $C < (m-1)/6$ needs further work. For $3 \leq m \leq 6$ the smallest retained unit cost $(m-1)/m$ already reaches this threshold, so no retained contact is possible. The literal $A_1$ case $m = 2$ is separate, and its modified target is zero. For $7 \leq m \leq 10$ a distance-two contact likewise reaches the threshold; only distance one remains. For $m \geq 10$ the original per-side budget two excludes every cyclic distance at least three, since $3(m-3)/m > 2$ and resistance increases up to the antipode. For a remaining distance $d = 1, 2$, polarization leaves the margin

$$\frac{4}{3}\frac{d(m-d)}{m} - \frac{d(m+\varepsilon)}{m} = \frac{d(m-4d-3\varepsilon)}{3m} \geq 0.$$

The nontrivial domain starts at $m = 7$ for $d = 1$ and $m = 11$ for $d = 2$ in odd parity, where the respective margins are zero; the even cases are nonnegative too. Smaller odd ranks with a negative formal margin were already in the trivial domain. The finite exceptional omission cases enumerate every cheap retained support and both omitted-node states. Single contacts give a convex quadratic whose endpoint and stationary minima are checked exactly; the sole cheap two-contact support has energy$(1 + u^2 + v^2)/2$. These finite certificates use the explicit grounded matrices above, so have no missing high-rank or finite-angle branch.

Summing and using(9) proves the second line of (11). The certificate verifies the stated finite quadratic and symbolic polynomial identities. Their domains, the contact-gap argument and the coordinate identities explain why they imply the all-rank inequality.

## E.7 A narrow facet without a cusp

For completeness we prove the compact-facet alternative rather than import a finer compact-chamber weighting. Let$Q$ be a compact two-face. Its white normals$W$ number$n - 2$, and every consecutive black pair with$W$ is positive definite. A positive-definite principal block gives an actual common face by the interior-point projection already proved. A singular positive-semidefinite block has a positive null combination in an affine component. It is nonzero, future null and nonpositive against every other chamber wall, so gives an ideal point of its face.

For a triangle, a minimal indefinite star subdiagram contains all three black nodes: omitting any one puts it in a spherical corner. By the order-ten classification it has at most seven whites. In a spanning tree joining the three marks, deleting either of two outer marks leaves the other two joined. The resulting two paths avoid the third mark and have at most eight edges, so lie in two actual corner diagrams.

For a quadrilateral split opposite black pairs as$U, V$. The block$W + U$ is indefinite. If positive definite, its actual face would make opposite sides meet. If positive semidefinite and singular, it would give an ideal point of the compact$Q$. Both are impossible. Likewise$W + V$ is indefinite. Minimal indefinite subdiagrams$L \subset W + U$, $M \subset W + V$ contain their respective black pairs and at most eight whites. They cannot be orthogonal, since their negative vectors would give negative index at least two. A shortest path in their overlapping or adjacent union between$U$ and$V$ has only white interior nodes and length at most$8 + 1 + 8 = 17$. Its black endpoints are consecutive sides, so the path belongs to an actual corner.

Thus unit weight at ambient spherical distance at most17 pays two on a compact triangle and one on a compact quadrilateral. Its cost at an ambient finite vertex is at most$17n$: spherical components of order at most35 have at most$\binom{r}{2} \le 17r$ pairs, and larger components are paths or$D$ forks with at most$r$ pairs at each fixed distance. This uses only the finite Coxeter classification, allowing affine subdiagrams elsewhere in the chamber.

A compact facet has dimension$d = n - 1$ and is simple, because its vertices are finite vertices of the Coxeter chamber. Its two-faces are compact faces of the ambient chamber; its angles count a subset of the ambient vertex pairs. Hence their total is at most $17(d + 1)f_0$. Its nonnegative symmetric$h$-vector has$h_0 = h_d = 1$. Pairing the coefficients$\binom{k}{2}$ and$\binom{d-k}{2}$ gives

$$\frac{f_2}{f_0} > \begin{cases} d(d-2)/8 & d \text{ even}, \\ (d-1)^2/8 & d \text{ odd}. \end{cases}$$

The inequality is strict because the nonzero endpoint coefficients exceed the middle minimum. Consequently the average required payment$5f_2/f_0 - \binom{d}{2}$ exceeds $d(d-6)/8$ or$(d-1)(d-5)/8$. After subtracting$17(d + 1)$, the differences at$d = 143 + 2t$ and$144 + 2t$ are respectively $t^2/2 + 36t + 3/2$ and$t^2/2 + 73t/2 + 19$, positive for$t \ge 0$. Thus a compact facet forces$n \le 143$.

A finite-volume facet with no ideal vertices is compact. Apply Nikulin's theorem and split by whether the chosen facet has a cusp. The cusp branch above excludes every$n \ge 164$, and the compact-facet branch already excludes$n \ge 144$. This completes the unbounded tail without assuming that the facet itself is a Coxeter polytope.

# F Minimal proof-atlas spine

| Role | Artifact stem |
|---|---|
| Final57 composition | `fresh_d13_context_dimension57_20261018` |
| Dimension-specific local rule | `fresh_d13_three_rank15_context_transfer_20261018` |
| B13 endpoint repair | `fresh_b13_endpoint_retention_20261018` |
| Complete preceding finite count | `fresh_d17_a16_coupled_ambient58_frontier_20261018` |
| Current coupled local rule | `fresh_d17_leaf_a16_context_coupled_transfer_20261018` |
| Positive low8 cusp certificate | `fresh_order10_low8_ambient58_frontier_20261018` |
| Genuine ideal consumer | `fresh_affine_d12_leaf_pair_transfer_20261017` |
| Triangle protection | `fresh_finite7_ideal5_triangle_payments_20261017` |

| | |
|---|---|
| Current finite6 interface | `fresh_depth6_finite6_zero_transfer_20261018` |
| Completed higher tail | `fresh_a15_d15_dimension58_20261018` |
| Ideal-five marked tables | `fresh_ideal5_small_factor_payments_20261017` |
| Ideal-six marked tables | `fresh_ideal6_combined_payments_20261017` |
| Finite rank17 marked tables | `fresh_finite_d17_long_pair_support_20261017` |
| Ordered residual model certificates | `fresh_nikulin_two_budget_frontier_20261015` |
| Independent four-thirds certificates | `fresh_nikulin_unit_charge_20261015_review` |
| Factor-fiber geometry | `fresh_cusp_factor_payment_framework_20261017` |
| Shape-curvature derivation | `fresh_cusp_link_shape_curvature_20261017` |

The full recursive manifests, not this compact index, identify all prior local inputs. The quantified map and reproduction guide are `docs/finite_volume57_lemma_map.md` and `docs/finite_volume57_reproduce.md`. The marked-table supplement uses records $(b, m, X, P)$ with literal pair lists, and preferred records with their labelled core overlap, marks and completion-order bound, as defined in Appendix D. The residual supplement records the exact model matrix, omissions, retained supports, KKT weights and slacks, domination record, and packing predecessors from Appendix E. These schemas specify what the certificate verifier checks; the artifact names in this index identify its retained data and provenance.

## Receipt for the finite layer tables

For clarity, the following receipt maps the mathematical table keys to exact array fields in the retained supplement. A row is selected by its canonical type and rank/order, never by its incidental array position. All listed stems below have suffix `_20261018.json` in the results directory. Complete weighted arrays and indicator arrays have the distinct defaults specified in Appendix D.6. The immutable composition manifest binds their exact bytes; this receipt is not an instruction to execute a producer.

| Mathematical table | Supplement stem | Exact array selector |
|---|---|---|
| $\mathsf{F}_{61,A,15}$ | `fresh_finite_a15_endpoint_coupon_zero` | `weighted_pairs` |
| $\mathsf{F}_{61,B/D,15}$ | `fresh_finite_b15_d15_existing_payment_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{F}_{61,A,16}$ | `fresh_finite_a16_endpoint_sparse_half_transfer` | `weighted_pairs` |
| $\mathsf{F}_{61,D,16/17}$ | `fresh_finite_d16_fork_exchange` | `D16_weighted_pairs`, `D17_weighted_pairs` |
| $\mathsf{F}_{60,A/B,15}$ | `fresh_rank15_sparse_payment_exchange` | `rules[kind].weighted_pairs` |
| $\mathsf{F}_{60,D,15}$ | `fresh_finite_d15_three_singleton_half_transfer` | `weighted_pairs` |
| Exchange base for $A_{16}, A_{17}$ | `fresh_a16_a17_full_payment_exchange` | `A16_weighted_pairs`, `A17_weighted_pairs`; then printed60 patch |
| Exchange base for $D_{16}, D_{17}$ | `fresh_d16_d17_eighth_balance` | `D16_weighted_pairs`, `D17_weighted_pairs`; then printed60 patch |
| $\mathsf{F}_{59,B/D,14}$ | `fresh_finite_rank14_existing_payment_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{F}_{59,A,14}$ | `fresh_finite_a14_near_endpoint_zero_transfer` | `weighted_pairs` |
| $\mathsf{F}_{59,A,15}$ | `fresh_finite_a15_three_coupon_zero_transfer` | `weighted_pairs` |
| $\mathsf{F}_{59,D,15}$ | `fresh_finite_d15_a4_a11_half_zero_transfer` | `weighted_pairs` |
| $\mathsf{F}_{59,D,16}$ | `fresh_finite_d16_alternate_half_zero_transfer` | `D16_weighted_pairs` |
| $\mathsf{F}_{59,A,17}$ | `fresh_a17_five_pair_alternate_transfer` | `weighted_pairs` |
| $\mathsf{I}_{65,m,X}$, $12 \le m \le 17$ | `fresh_deep_finite8_dimension64` | `short7_small_factor_interfaces[order,kind].weighted_pairs` |
| $\mathsf{I}_{60,12,B/C}$ | `fresh_order12_bc_near_terminal_half_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{I}_{60,13,B/C}$ | `fresh_order13_bc_sparse_terminal_transfer` | `records[kind].weighted_pairs` |

| Mathematical table | Supplement stem | Exact array selector |
|---|---|---|
| $\mathsf{I}_{60,13,D}$ | `fresh_order13_d_center_leaf_half_transfer` | `weighted_pairs`; absent pairs zero |
| $\mathsf{I}_{60,14,C}$ | `fresh_order14_c_distance10_terminal_projection` | `weighted_pairs` |
| $\mathsf{I}_{60,15,B/D}$ | `fresh_order15_bd_terminal_coupon_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{I}_{60,16,B/D}$ | `fresh_order16_bd_sparse_unit_transfer` | `records[kind].weighted_pairs` |
| $\mathsf{I}_{60,17,D}$ | `fresh_small_d16_coupon_and_ideal17_rebalance` | `local_proofs.weighted_pairs` |
| $\mathsf{I}_{59,13,B/C}$ | `fresh_order13_bc_inner_terminal_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{I}_{59,14,B}$ | `fresh_order14_b_basic_distance9_transfer` | `weighted_pairs` |
| $\mathsf{I}_{59,14,C}$ | `fresh_order14_c_basic_distance9_transfer` | `weighted_pairs` |
| $\mathsf{I}_{59,14,D}$ | `fresh_order14_d_near_fork_half_transfer` | `weighted_pairs` |
| $\mathsf{I}_{59,15,B/D}$ | `fresh_order15_bd_basic_fractional_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{I}_{59,16,C/D}$ | `fresh_order16_cd_sparse_sixteenth_transfer` | `rules[kind].weighted_pairs` |
| $\mathsf{I}_{59,17,D}$ | `fresh_order17_center_contextual_zero_transfer` | `local_proof.weighted_pairs` |

For the preferred tables $P_b$ the retained source arrays are `order17.targets[kind].paid_pairs` in the ideal-seven table; `targets[kind].paid_pairs` in the order16 preferred table, with the nine printed$D$ removals; `small_order_pair_sets[order,kind].paid_pairs` in `fresh_ideal6_combined_payments_20261017.json`; and `records[order,kind].paid_pairs` in the ideal-five table. The rank17 finite support table uses `records[kind].retained_pairs`. Their mathematical graphs, domains, induced-map algorithm and default values were defined before this provenance index.

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